Exploring the Differential Impact of Repealing the ACA Individual Mandate by Health Status[†]

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Abstract

Health insurance mandates are designed to mitigate adverse selection by encouraging enrollment among relatively healthy individuals. Yet, little is known about how removing these incentives affects relatively healthy and sick consumers. This paper examines the differential impact of repealing the Affordable Care Act (ACA) individual mandate on insurance participation, Marketplace premiums, out-of-pocket premium spending, and financial incidence. I exploit cross-state variation in exposure to the elimination of the individual mandate in 2019 using a two-way fixed effects (TWFE) design that compares states without an individual mandate in effect with Massachusetts, New Jersey, and the District of Columbia, that maintained mandate enforcement. I find that the repeal increased the probability of becoming newly uninsured by 0.56 percentage points (approximately 27 percent relative to the pre-reform mean), with a statistically significant increase among relatively healthy consumers, but no statistically significant change among relatively sick consumers. The benchmark Marketplace premiums for a 27-year-old enrollee increased by approximately $739, while average out-of-pocket premium spending declined by $284.60. This divergence is consistent with ACA premium subsidies limiting eligible consumers' exposure to increases in gross Marketplace premiums. The findings highlight how the ACA subsidy structure may alter the distributional consequences of adverse selection by limiting the extent to which higher Marketplace premiums are passed through to eligible consumers.

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1 Introduction

A major rationale for insurance mandate is its potential, in conjunction with other policies, to help mitigate adverse selection in insurance markets (Akerlof, 1970; Rothschild and Stiglitz, 1976; Wilson, 1977). However, critics argue that mandating coverage constrains individual choice and imposes financial burdens, particularly on low-income individuals (Young, 2009; Lurie et al., 2021; Soni, 2022). The absence of a mandate likewise has consequences. Proponents of repealing the mandate contend that it relieves low-risk individuals of the premium costs associated with being insured or of tax penalties for not having coverage. In contrast, opponents argue that removing the mandate can lead to an increasingly high-risk insurance pool, placing upward pressure on premiums and healthcare expenditures (Saltzman, 2019). In addition, those who drop coverage are now exposed to more financial risk due to an unexpected health care expense.

Despite these debates, relatively little is known about how repealing insurance mandates affects consumer coverage decisions and healthcare spending by health status. In this paper, I estimate the impact of the 2019 repeal of the Affordable Care Act (ACA) individual mandate on insurance coverage and expenditure. If the insurance mandate imposes financial costs on individuals with relatively low demand for coverage, then its repeal may yield short-term financial gains for them as they drop coverage. On the other hand, increased exposure to health shocks and out-of-pocket financial risk may offset those gains to some extent. In addition, those with relatively high demand for coverage (i.e. the sick) now face higher premiums due to adverse selection. Understanding how the removal of insurance mandates affects consumers with different levels of health status is therefore important for both economists and policymakers interested in designing welfare-improving insurance policies.

Before the repeal, Americans with household incomes above 138% of the federal poverty line were required to obtain coverage or pay a tax penalty. Previous work attempting to isolate the effect of the mandate apart from the other components of the ACA found that the mandate increased insurance coverage by roughly 2.5 percentage points in 2016 (Lurie et al., 2021). However, the ACA individual mandate was repealed by the "Tax Cuts and Jobs Act" of 2017, which set the penalty for not having health insurance to zero dollars, effective from January 1, 2019. After the federal individual mandate penalty was reduced to zero in January 2019, Massachusetts retained its existing state individual mandate, while New Jersey and the District of Columbia implemented state-level mandates, leaving 48 states without an individual mandate in effect (Soni, 2022).[1] This cross-state variation in mandate enforcement provides a natural experiment to study how its elimination shapes insurance coverage and healthcare spending by health status.

To guide my empirical analysis, I build on the conceptual framework developed by Geruso et al. (2023a), which distinguishes between insurance demand at the extensive margin (whether to purchase insurance) and the intensive margin (the generosity of insurance coverage). Their framework predicts that changes in mandate policy can affect both the decision to purchase insurance and the choice of insurance generosity, with implications for equilibrium premiums and welfare. My analysis applies this framework to the repeal of the ACA individual mandate in order to generate testable predictions about changes in insurance coverage and out of pocket premium payments driven by the repeal.

I test these predictions using a sample of non-elderly adults from the 2015–2019 Current Population Survey Annual Social and Economic Supplement (CPS ASEC). The CPS ASEC reports both current and past-year insurance coverage, allowing me to examine how consumers adjusted their coverage decisions following the repeal. It also contains information on out-of-pocket premium spending and self-reported health status, enabling me to estimate how the repeal affected insurance coverage and expenditure by health status.

My analysis addresses three related questions. First, I examine whether the repeal of the ACA individual mandate increased adverse selection. I find that repeal increased the probability of becoming newly uninsured by 0.56 percentage points which is approximately 27% relative to pre-reform mean among the full sample. The increase is statistically significant among relatively healthy individuals, while the estimated effect among relatively sick individuals is not statistically distinguishable from zero.

[1] Figure 1 presents the variation in insurance mandate penalty after the repeal in 2019

Second, I study the effect of the repeal on consumers' out-of-pocket premium expenditure. I find that out-of-pocket premium spending declined by $284.60 (approximately 18%) relative to the pre-reform mean for the full sample. Again, I find a statistically significant decline in out-of-pocket premium spending among relatively healthy consumers. Although I find a similar decrease among higher-risk consumers, the estimate is not statistically distinguishable from zero.

Third, I estimate the impact of the repeal on benchmark Marketplace premium. I find that the repeal increased the annual benchmark Marketplace premium by $738.93 for a 27-year-old individual in 2019 equivalent to approximately 21% of the pre-repeal mean. The corresponding increase was $1,361.46 for a 50-year-old individual and $3,341.60 for a 30-year-old couple. These estimates indicate that the repeal substantially increased the unsubsidized (i.e. gross) cost of insurance in the individual market. The larger dollar increases for older adults and couples are consistent with the ACA's age-rating structure and their higher expected medical costs.

This paper makes two contributions to the literature. First, it contributes to the growing literature on the effects of the ACA individual mandate repeal on insurance coverage (Zhang, 2021; Soni, 2022). While existing studies document that repealing the mandate increases the uninsured rate, they provide limited evidence on which consumers leave the insurance market. I fill this gap by showing that the increase in the newly uninsured is concentrated among relatively healthy consumers, whereas coverage among relatively sick consumers remains largely unchanged.

Second, this paper contributes to the literature on adverse selection in health insurance markets (Einav et al., 2010; Einav and Finkelstein, 2011; Castaneda and Marton, 2013).[2] While this literature has shown that adverse selection raises equilibrium premiums and reduces welfare, there is comparatively little causal evidence on how these market-level effects translate into consumers' out-of-pocket premium spending. Distinguishing between equilibrium premiums and the premiums consumers actually pay is particularly important in the presence of premium subsidies, which can insulate insured individuals from increases in market premiums. I provide causal

[2]For a more recent literature, see Handel et al. (2015), Hackmann et al. (2015), Geruso and Layton (2017), Marton et al. (2017), Saltzman (2018), Saltzman (2019), Lurie et al. (2021), Shepard (2022), Geruso et al. (2023a), Geruso et al. (2023b), Kreider et al. (2024), and Peterson (2025).

evidence that, although the repeal of the ACA individual mandate induced the exit of relatively healthy consumers and intensified adverse selection, out-of-pocket premium payments declined rather than increased. I argue that this pattern reflects the ACA's premium subsidy structure, which limits premium pass-through for eligible households with incomes between 138% and 400% of the federal poverty line. More broadly, my findings show that premium subsidies can substantially attenuate the extent to which increases in equilibrium premiums are passed through to consumers following mandate repeal.

The rest of the paper is structured as follows. Section 2 focuses on the conceptual framework describing the interaction between adverse selection, individual mandate penalty and its repeal. In Section 3, I provide the background on my empirical setting from the ACA individual mandate. Section 4 describes the data, and Section 5 discusses the empirical strategy. Section 6 presents the results, and Section 7 concludes the paper.

2 Conceptual Framework

I begin by adapting the conceptual framework of Geruso et al. (2023a) to illustrate how an individual mandate affects insurance demand in a market with adverse selection. Their model allows consumers to choose among a more generous plan, Q_H, a less generous plan, Q_L, and an outside option of remaining uninsured, Q_U.

Imposing a mandate penalty raises the cost of Q_U and induces relatively healthy, low-cost consumers at the $Q_L - Q_U$ margin to enroll in Q_L. Their entry lowers the average cost, and hence the equilibrium price of Q_L. The resulting reduction in the price of Q_L then induces some relatively higher-risk consumers at the $Q_H - Q_L$ margin to switch from Q_H to Q_L, shifting demand away from the more generous plan, as illustrated in Figure 2. Thus, the mandate generates responses along both the extensive and intensive margins: it increases insurance coverage among relatively healthy consumers while inducing some higher-risk consumers to substitute toward less generous coverage. These cross-margin responses imply that the effects of a mandate cannot be inferred from its direct effect on insurance participation alone.

These effects operate in reverse when the mandate penalty is removed. Relatively healthy, low-cost consumers at the $Q_L - Q_U$ margin move from plan Q_L into uninsurance, worsening the risk composition of plan Q_L and raising its equilibrium price. The resulting increase in the price of Q_L induces some relatively higher-risk consumers at the $Q_H - Q_L$ margin to switch toward the more generous plan Q_H. This intensive-margin response changes the allocation of higher-risk consumers across plans and thus the equilibrium price of each plan.

This framework motivates my empirical analysis as it yields several testable predictions. Following the removal of the mandate, relatively healthy consumers on the margin of having coverage or not should be more likely to return to non-coverage. Higher-risk consumers are predicted to remain insured in the more generous plan. Appendix A.1 provides a more detailed illustration of these predictions using the institutional settings of the ACA individual mandate. The effect on out-of-pocket premium spending among relatively healthy consumers is ambiguous. Lower-risk consumers who remain insured may face higher equilibrium premiums in the less generous plan if some healthy consumer drop coverage, however those premiums could also fall as consumers on the margin between the more and less generous plans move back into the more generous plan. In addition, those who become uninsured no longer make premium payments. Higher-risk consumers, who are more likely to remain insured in the more generous plan, may face slightly lower premiums due to consumers on the margin moving from the less generous to the more generous plan.

3 Background on the ACA Individual Mandate

The Affordable Care Act (ACA), signed into law in March 2010, represents the largest overhaul of the U.S. healthcare system since the launch of Medicare and Medicaid in the mid-1960s (Frean et al., 2017). It was primarily designed to expand health insurance coverage in the United States by implementing market reforms, individual mandates, premium subsidies, health insurance exchanges, and Medicaid expansions, while also aiming to address the escalating cost of healthcare (Gruber, 2011; Courtemanche et al., 2017).

The major components of the ACA, including the individual mandate, were introduced in January 2014. The mandate requires Americans with household incomes above 138% of the FPL to buy health coverage or face a tax penalty (Gruber, 2011; Courtemanche et al., 2017).[3] The goal of the ACA individual mandate was to mitigate adverse selection by encouraging healthier individuals to buy coverage (Fiedler, 2020; Soni, 2022). However, exemptions were granted to specific groups, including individuals with an income below the tax filing threshold and those who did not have access to a health insurance plan costing less than 8% of their annual income in 2014 and 8.05% in 2015 (Saltzman, 2019). In addition, exemptions were granted to those who would have qualified for Medicaid but lived in a non-expansion state (Saltzman, 2019).

Critics of the mandate provision consider it an undue infringement on their freedom and a financial burden on low-income Americans in spite of the subsidies built into the ACA (Young, 2009; Lurie et al., 2021; Soni, 2022). In 2017, the United States Congress passed the "Tax Cuts and Jobs Act", which effectively repealed the federal individual mandate by reducing the penalty for not having health insurance to zero dollars, starting from January 1, 2019. After the federal mandate penalty was reduced to zero, Massachusetts retained its pre-existing state individual mandate, while New Jersey and the District of Columbia implemented state-level mandates effective in 2019.[4] Thus, uninsured individuals living in states with an individual mandate are still penalized, while uninsured residents in states without such a mandate are not penalized.[5]

[3]In 2014, the penalty was the greater of 1% of household income, capped at the national average annual premium for a Bronze plan, or a flat fee of $95 per adult and $47.5 per child, with a maximum of $285 per family. By 2015, the penalty was the greater of either $325 per adult or 2% of household income capped at $975 per family; and in 2016, the penalty was the greater of either $695 per adult or 2.5% of household income capped at $2085 (Courtemanche et al., 2017; Saltzman, 2018).

[4]The following year, two additional states (California, and Rhode Island) joined the list of states with health insurance mandates (Soni, 2022).

[5]Table B.1 in the appendix summarizes states with and without mandates in 2019.

4 Data Description

I combine several data sources to estimate the effects of the ACA individual mandate repeal. I use the Current Population Survey Annual Social and Economic Supplement (CPS ASEC) to measure insurance coverage and consumers' out-of-pocket premium payments and the Robert Wood Johnson Foundation (RWJF) Health Insurance Exchange data to measure market premiums. In supplementary analysis, I use the Behavioral Risk Factor Surveillance System (BRFSS) to examine changes in healthcare utilization. Together, these data sources allow me to study the effects of mandate repeal on insurance participation, market premiums, consumer premium payments, and healthcare utilization by health status.

For my main analysis, I use data from the 2015–2019 CPS ASEC (Flood et al., 2023). Conducted annually between February and April, the CPS ASEC provides detailed information on respondents' health insurance coverage, state of residence, and socioeconomic characteristics. Importantly, the survey reports both current and prior-year insurance coverage, allowing me to identify individuals who were previously insured but are uninsured at the time of the interview. This feature enables me to examine changes in insurance coverage following the repeal of the ACA individual mandate.

I construct "Newly Uninsured", an indicator equal to one if a respondent was insured in the previous year but uninsured at the time of the interview, and zero otherwise. Similarly, I define "Newly Insured" as an indicator equal to one if a respondent was uninsured in the previous year but insured at the time of the interview, and zero otherwise.[6] I also construct an indicator for "good health" using respondents' self-reported health at the time of the survey interview. The indicator equals one for respondents reporting excellent, very good, or good health and zero for those reporting fair or poor health. I refer to these groups as relatively healthy and sick consumers, respectively. The 2015–2019 CPS ASEC also reports respondents' out-of-pocket premium payments, which I use to estimate the effect of mandate repeal on consumer premium spending by health status. Finally, the survey provides detailed demographic and socioeconomic character-

[6] I use these measures to validate the findings of Soni (2022) (see Appendix Table B.1).

istics, including state of residence, age, sex, marital status, educational attainment, employment status, and race and ethnicity.

There are many advantages associated with using the CPS ASEC data. First, it is less prone to insurance misreporting. Instead of asking respondents a series of yes/no questions, the survey first asks whether they have coverage, then asks about their general sources of insurance, and finally asks follow-up questions about their specific coverage (Pascale et al., 2019). Second, the CPS-ASEC survey on insurance measures has been independently validated as more accurate compared to other notable health insurance surveys, which tend to overestimate the uninsured population (Pascale et al., 2019; Soni, 2022). These features give confidence in the reliability of the insurance measures used in my analysis.

In addition, I use 2015–2019 Health Insurance Exchange data from the RWJF to examine how mandate repeal affected market premiums. This dataset includes detailed information on premiums offered by insurers across the 50 states and the District of Columbia, including state-year benchmark premiums before the application of ACA premium subsidies. These data allow me to distinguish between changes in market premiums from changes in the out-of-pocket premium payments observed in the CPS ASEC. The two premium measures capture different concepts and populations. The RWJF measure is the gross benchmark premium offered in the ACA Marketplace before premium subsidies, whereas CPS ASEC out-of-pocket premium spending reflects household premium payments across insurance sources and is not limited to Marketplace coverage. Thus, the two measures should not be interpreted as gross and net prices for the same insurance contract or the same population. I use the RWJF data to examine changes in Marketplace benchmark premiums and the CPS ASEC to examine changes in consumers' reported out-of-pocket premium spending.

5 Empirical Strategy

This section describes the empirical methodology used throughout the paper. I begin by outlining the identification strategy used to estimate the causal effects of the repeal of the ACA individual mandate. I then present the two-way fixed effects model used to estimate the effects on the outcomes of interest, followed by a two-part model for outcomes with a large share of zero observations. Finally, I discuss the robustness checks and identification assumptions used to assess the validity of the empirical estimates.

5.1 Identification Strategy

To identify the causal effects of mandate repeal on insurance coverage and out-of-pocket premium spending by health status, I restrict the sample to non-elderly adults with household incomes between 138% and 400% of the FPL. I exclude adults aged 65 and older because Medicare eligibility largely insulates them from the effects of the individual mandate. Individuals younger than 19 are excluded because health insurance decisions for this group are typically made by parents or guardians rather than by the individuals themselves. I also exclude individuals with household incomes below 138 percent of the FPL because they are generally eligible for Medicaid and therefore face different insurance incentives than those purchasing private coverage.

Finally, I exclude individuals with household incomes above 400% of the FPL because they are generally ineligible for ACA Marketplace premium subsidies during the study period. Moreover, higher-income households are more likely to obtain coverage through employer-sponsored insurance rather than the individual Marketplace. I restrict the baseline sample to adults with household incomes between 138 and 400 percent of the federal poverty line, the income range that generally encompassed eligibility for ACA premium tax credits during the study period, subject to additional eligibility requirements.

5.2 Two-Way Fixed Effects

I exploit cross-state variation in exposure to the elimination of an enforceable individual mandate in 2019. Following the reduction of the federal mandate penalty to zero, individuals residing in states without state-level mandates were no longer subject to an enforceable individual mandate, whereas individuals in Massachusetts, New Jersey, and the District of Columbia remained subject to mandate enforcement. I estimate the following baseline model using the two-way fixed effects (TWFE) framework:

$$\mathrm{y}_{ist}^{k} = \alpha + \beta 1[\text{Treatment}_s \times \text{post2019}_t] + \gamma X_{ist} + \lambda_s + \tau_t + \epsilon_{ist} \tag{1}$$

where y_{ist}^{k} denotes the outcome for individual *i* living in state *s* in year *t*. The outcomes include out-of-pocket premium spending, the probability of becoming newly uninsured, and the probability of becoming newly insured. $Treatment_s$ is an indicator equal to one for individuals residing in states without an individual mandate in effect in 2019 and zero for individuals residing in jurisdictions with an individual mandate in effect in 2019. $post2019_t$ is an indicator equal to one for observations in 2019 and later, and zero otherwise. X_{ist} is a vector of individual demographic and socioeconomic characteristics, including age, sex, race and ethnicity, marital status, educational attainment, household size, and employment status. The model also includes state fixed effects, λ_s, and year fixed effects, τ_t. Finally, ϵ_{ist} denotes the idiosyncratic error term.

Using the TWFE framework, I control for individuals' state of residence to account for unobserved differences across states. I also control for the year of the interview to account for variations in nationwide trends in insurance coverage. The parameter β captures the differential change in outcomes after 2019 between states without an individual mandate in effect and jurisdictions that maintained mandate enforcement. The estimates include CPS ASEC survey weights and standard errors are clustered at the state level.

5.3 Two-Part Model

Similar to other healthcare spending data, out-of-pocket premium spending for health insurance coverage is skewed because (i) out-of-pocket premium spending cannot be negative, (ii) there is a concentration of values at zero for both insured and uninsured individuals, and (iii) older adults are likely to pay higher premiums than younger adults. Given the presence of mass zeroes in the out-of-pocket premium spending distribution, I employ a two-part model approach developed by Belotti et al. (2015) to estimate the effects of repealing the ACA individual mandate on out-of-pocket premium spending. This model effectively addresses the skewness associated with discrete-continuous outcomes, which is a common feature of medical expenditure data. It applies the basic principles of probability to estimate the parameters in the conditional mean function $E(Y|X) = (Y > 0|X) \times E(Y|Y > 0, X)$.

The first part of the model estimates the probability of positive out-of-pocket premium-spending. In other words, the outcome represents having any spending vs none. I use the logit model to estimate the probability of incurring out-of-pocket premium spending $pr(Y_{iast} > 0|x) = \frac{1}{1+e^{-\beta X}}$. Where Y_{ist} is a binary outcome equal to 1 if out-of-pocket premium spending for an individual i residing in state s in year t is positive and zero otherwise. Here, X is a vector of covariates that includes the interaction term.

In the second part, I use the Generalized Linear Model (GLM) developed by McCullagh (2019) to model the out-of-pocket premium spending conditional on positive spending. Unlike OLS, which often requires transformations of the dependent variable, the GLM accommodates the skewed distribution and heteroskedasticity commonly observed in expenditure data (Manning and Mullahy, 2001). Therefore, I specify a GLM with a log-link function and gamma distribution to estimate the effect of the repeal on out-of-pocket premium spending among individuals with positive spending: $E(Y_{ist}|Y_{ist} > 0, X) = e^{X\theta}$. Where Y_{ist} denotes the out-of-pocket premium spending incurred by individual i residing in state s in year t, and X is a vector of covariates as specified in the first part of the analysis.

I combine the two components by computing the predicted unconditional out-of-pocket premium spending for each observation as the product of the predicted probability of positive expenditure from the first-part logit model and the predicted conditional expenditure from the second-part GLM. I then calculate the average marginal effect of the treatment-by-post interaction on unconditional expenditure. Thus, the estimates reported in Table 4 are expressed in annual U.S. dollars rather than as coefficients from either component of the two-part model. Standard errors for the combined average marginal effects are obtained using the delta method and are clustered at the state level.

5.4 Validation of Identifying Assumptions

The idea behind the TWFE model, as expressed in Equation 1, is based on two key assumptions. First, my outcome variables of interest would have trended similarly in the treatment and control states in the absence of the policy change. Although I cannot test the counterfactual directly, I can compare trends in the pre-period by estimating Equation (2). The absence of differential trends in the pre-period suggests evidence that my identifying assumptions hold.

$$\mathrm{y}_{ist}^{k} = \alpha + \sum_{\substack{j=2015 \\ j\neq\{2018\}}}^{2019} \beta_j 1[\mathrm{Treatment}_{sj}] + \gamma X_{ist} + \lambda_s + \tau_t + \epsilon_{ist} \tag{2}$$

Where $Treatment_{sj}$ is a set of event-time indicators identifying the years before and after the repeal of the federal individual mandate. The event-time variable, *j*, indexes years relative to the repeal, with 2018 serving as the omitted reference year. Values of *j* after 2018 represent post-repeal periods (lags), whereas values before 2018 represent pre-repeal periods (leads). The coefficient β_j measures the effect of mandate repeal on the outcome of interest relative to the reference year. All other variables are defined as in Equation (1).

Second, I assume that the outcomes in both the treatment and control states should not undergo any further differential changes during the post-treatment period. The validity of this assumption was tested by conducting two falsification tests using Equation (1) across two groups of adults with no incentive to drop insurance coverage following the mandate repeal. I regress

Equation (1) on the elderly adults in my sample (who were eligible for Medicare before and after the repeal and are unaffected by the mandate repeal). I also show the estimates for low-income individuals with household incomes below 138% of the federal poverty line (who are eligible for Medicaid and were exempted from the ACA mandate penalty before and after the policy change). These groups are less likely to be affected by the mandate repeal. Therefore, a significant causal effect of the repeal on my outcome variables of interest for either of these two groups would raise serious concerns about the interpretation of my primary results if there are any simultaneous changes in other policies.

I conduct several robustness checks to assess the validity of the main results. First, I re-estimate Equation (1) using a broader sample that includes individuals with household incomes below 138% of the FPL. Second, because the treatment group is substantially larger and more geographically diverse than the control group (Massachusetts, New Jersey, and the District of Columbia), I restrict the sample to Northeastern states to improve the comparability of treated and control states. Finally, I re-estimate Equation (1) excluding observations from 2018. This robustness check accounts for the possibility that the announcement of the Tax Cuts and Jobs Act in late 2017 may have affected insurance decisions before the federal mandate penalty was reduced to zero in 2019 (Fung et al., 2019; Soni, 2022).

Finally, I examine whether the effects of mandate repeal vary across population subgroups by estimating the model separately for men and women, parents and childless adults, married and unmarried individuals, age groups, employment status, and educational attainment. I also conduct a series of placebo tests using outcomes that should not be directly affected by the repeal, including labor force participation, and self-employment. Finding no significant effects on these placebo outcomes would strengthen the interpretation that the estimated effects reflect changes induced by mandate repeal rather than unobserved confounding factors.

6 Results and Discussion

In this section, I present my empirical findings. Using both graphs and regression analysis, I show the impact of repealing the ACA individual mandate on insurance coverage, and out-of-pocket spending, and Marketplace premiums.

6.1 Baseline Estimates of the Effects of Repealing the ACA Mandate

I begin by presenting descriptive statistics for the 2015–2019 CPS ASEC sample in Table 1. The table reports the means of the outcome variables and respondents' demographic characteristics before and after the repeal of the ACA individual mandate for both the treatment and control states. The demographic characteristics are broadly similar across the two groups prior to the repeal, suggesting that the treatment and control samples are comparable along observable characteristics. Column 5 reports the unadjusted difference-in-differences in sample means after the repeal.

Figure 3 plots trends in individual Marketplace enrollment using open enrollment data from the Centers for Medicare and Medicaid Services (CMS).[7] The solid blue line shows total Marketplace enrollment, and the vertical line marks the repeal of the ACA individual mandate. Enrollment declines very modestly between 2018 and 2020, a pattern that is consistent with the repeal of the individual mandate following the Tax Cuts and Jobs Act of 2017. Enrollment begins to recover after 2020, likely reflecting expanded enrollment initiatives and temporary policy changes introduced during the COVID-19 pandemic. For completeness, the figure presents enrollment trends from 2015 through 2026. However, the empirical analysis is restricted to 2015 through 2019 to avoid the change in the way the CPS measured insurance between 2014 and 2015 and the potential confounding effects associated with the COVID-19 pandemic starting in 2020.

Table 2 reports the estimated effects of repealing the ACA individual mandate on health insurance coverage using Equation (1) for the full sample and separately for the healthy and sick

[7] CMS reports annual open enrollment statistics for the federal Marketplace and state-based health insurance exchanges.

sub-samples. Columns 1–3 present the estimated effects on the probability of becoming newly uninsured. All specifications include individual demographic and socioeconomic controls. The repeal increased the probability of becoming newly uninsured by 0.56 percentage points ($p < 0.01$) in the full sample, equivalent to approximately 27% of the pre-reform mean. This estimate is close to the increase of 0.5 percentage points reported by Soni (2022). Stratifying the sample by health status shows that the repeal increased the probability of becoming newly uninsured by 0.57 percentage points among relatively healthy consumers. By contrast, I do not find statistically significant change among sick consumers.

The subgroup estimates are consistent with the adverse-selection mechanism: the estimated coverage response is statistically significant among relatively healthy consumers but not among relatively sick consumers. However, the difference between the subgroup estimates is not statistically significant. These findings are consistent with the conceptual framework developed in Section 2, although the lack of a statistically significant difference across health-status groups limits the strength of the evidence for differential responses by health status.

Columns 4–6 report the estimated effects of mandate repeal on the probability of becoming newly insured. The repeal reduced the probability of becoming newly insured by 0.79 percentage points ($p < 0.01$), equivalent to approximately 34% of the pre-reform mean. This estimate is consistent with previous evidence that repealing the ACA individual mandate reduced insurance participation among individuals who would otherwise have enrolled under the mandate (Soni, 2022). Again stratifying the sample by health status shows that the decline in new insurance enrollment is concentrated among relatively healthy consumers, who became less likely to transition into coverage following the repeal. By contrast, I find no statistically significant effect among sick consumers, consistent with their greater willingness to pay for health insurance. Together with the results for newly uninsured individuals, these findings show that the estimated changes in insurance transitions are statistically significant among relatively healthy consumers but not among relatively sick consumers. However, formal tests do not reject equality of the treatment effects across health-status groups. The subgroup estimates therefore provide suggestive, rather than conclusive, evidence of differential responses by health status.

Table 3 examines whether the health composition of the insured population changed following the repeal. I find that the share of insured individuals who were relatively sick increased by 1.3 percentage points in states without a mandate, while the share who were relatively healthy declined by the same amount. Compared with the pre-repeal sick share of 12 percent, this represents about a 10.8 percent increase. Although these estimates are not statistically significant, the point estimates are consistent with the deterioration in the health composition of the insured pool after the repeal.

Table 4 reports the effects of mandate repeal on consumers' out-of-pocket premium spending using Equation (1). Columns 1–3 present the estimates for the full sample. The repeal reduced annual out-of-pocket premium spending by \$284.60, equivalent to approximately 18% of the pre-reform mean. Stratifying the sample by health status shows that thi decline is statistically significant among relatively healthy consumers, whose annual premium spending fell by \$282.59. Although out-of-pocket premium spending also declined among sick consumers, the estimate in Column 3 is not statistically distinguishable from zero. Columns 4–6 restrict the sample to respondents with current health insurance coverage. The results are similar to those for the full sample. Among insured individuals, annual out-of-pocket premium spending declined by \$282.76 overall and by \$286.18 among relatively healthy consumers. The estimated decline of \$186.92 among relatively sick consumers is smaller in magnitude and statistically indistinguishable from zero.

The conceptual framework developed in Section 2 yields different predictions for out-of-pocket premium spending across health-risk groups. For relatively healthy consumers, the overall effect is theoretically ambiguous. Following mandate repeal, the healthiest enrollees exit the insurance market and no longer pay insurance premiums. At the same time, the deterioration of the insured risk pool raises equilibrium premiums for those who remain enrolled, while changes in plan choice partially offset these increases. The estimates in Table 4 indicate that the net effect is negative, with relatively healthy consumers experiencing a significant decline in out-of-pocket premium spending.

For relatively sick consumers, the conceptual framework predicts a modest decline in out-of-pocket premium spending as some higher-risk enrollees shift toward more generous coverage, improving the risk composition of the less generous plan. Consistent with this prediction, I estimate a reduction in out-of-pocket premium spending among sick consumers. However, I find that the estimate is not statistically significant, suggesting that mandate repeal had little measurable effect on the premiums paid by consumers with greater healthcare needs.

Next, Columns 1–3 of Table 5 report the estimated effects of repealing the ACA individual mandate on Marketplace premiums using Equation (1). The estimates show that the repeal increased the annual bench mark premium by \$738.93 for a 27-year-old individual in 2019 equivalent to approximately 21% of the pre-repeal mean. The corresponding increase was \$1,361.46 for a 50-year-old individual and \$3,341.60 for a 30-year-old couple. These estimates indicate that the repeal substantially increased the unsubsidized (i.e. gross) cost of insurance in the individual market. The larger dollar increases for older adults and couples are consistent with the ACA's age-rating structure and their higher expected medical costs. Overall, the results suggest that the exit of relatively healthy consumers following mandate repeal raised equilibrium premiums, consistent with the adverse-selection mechanism developed in the conceptual framework.

The results document parallel but distinct reduced-form responses: gross Marketplace benchmark premiums increased, while average reported out-of-pocket premium spending declined. This divergence is consistent with the ACA subsidy structure limiting eligible consumers' exposure to increases in gross premiums.

6.2 Event Study

I estimate event-study specifications for all outcomes using Equation (2). Figure 4 reports the event-study estimates for the probabilities of becoming newly insured (Panel A) and newly uninsured (Panel B) using the 2015–2019 CPS ASEC. The estimates are presented separately for the full sample and for the healthy and sick sub-samples. All coefficients are normalized to zero in 2018, the year immediately preceding the repeal, so that each estimate is interpreted relative to the pre-repeal period. Across all samples, the pre-repeal coefficients are economically small and statistically indistinguishable from zero, indicating no evidence of differential pre-trends between the treatment and control states. These findings support the parallel-trends assumption and reinforce the validity of the empirical strategy.

Figure 5 presents the event-study estimates for out-of-pocket premium spending before and after the repeal of the ACA individual mandate. Panel A reports estimates for the full sample, including both insured and uninsured individuals, while Panel B restricts the sample to individuals with insurance coverage. Within each panel, estimates are reported for the overall sample and separately for relatively healthy and sick individuals. The coefficients are normalized to zero in 2018, the year immediately preceding the repeal. Across both panels and health-status groups, the pre-repeal estimates are small and statistically indistinguishable from zero, providing no evidence of differential pre-trends between treatment and control states. These patterns support the parallel-trends assumption underlying the empirical design used in this paper.

6.3 Heterogeneity in the Repeal of Individual Mandate Reform

Table 6 reports heterogeneity analyses for relatively healthy consumers using the CPS ASEC. Columns 1–2 present estimates for insurance transitions. I examine whether the effects of mandate repeal vary by sex, age, employment status, marital status, and parental status. I find that the point estimates vary across demographic groups. The largest responses are observed among unmarried individuals, childless adults, the self-employed, younger adults ages 19–26, and adults ages 60–64. These patterns are consistent with differences in insurance demand and reliance on the individual insurance market. By contrast, I find little evidence of differential responses among adults aged 27–59. Overall, these patterns provide suggestive evidence of heterogeneous responses across demographic groups.

Column 3 reports heterogeneity in the effects of mandate repeal on out-of-pocket premium spending. The decline in premium spending is larger among women, married individuals, adults with children, and workers who are not self-employed, as well as among adults ages 27–49. One possible explanation is that some consumers responded to the repeal by switching to less generous or cheaper premium plans. In addition, ACA premium subsidies may have protected eligible enrollees from increases in benchmark Marketplace premiums.

Table 7 examines heterogeneity in out-of-pocket premium spending by household income. Premium spending declines among individuals with household incomes below 250% of the federal poverty line, although the estimate for those between 138 and 150% is not statistically significant. These estimates are consistent with the ACA premium subsidy structure limiting eligible consumers' exposure to increase in benchmark premiums. The estimate for individuals between 250 and 300% of the FPL is positive but statistically indistinguishable from zero, while premium spending declines substantially among individuals between 300 and 400% of the FPL. Overall, the income-specific estimates provide suggestive evidence consistent with the subsidy mechanism, but the effects are not monotonic across income groups. Because income is also correlated with insurance source, employment, plan choice, and other determinants of premium spending, these estimates should not be interpreted as identifying the causal effect of subsidy mechanism.

6.4 Robustness Checks

The main analysis in this paper relies on two core assumptions of the standard difference-in-differences approach: that treatment and control groups followed parallel trends before 2019, and that individuals did not change their behavior in anticipation of the mandate repeal. To address potential concerns about these assumptions, I conducted a series of robustness checks, including alternative model specifications, sensitivity analyses, placebo tests, and falsification tests. These exercises assess the sensitivity of the main findings to alternative specifications and potential threats to identification.

In this section, I begin with Appendix Figure B.1, which plots trends for newly uninsured, premium spending, and medical out-of-pocket expenditure between the treatment and control states using the CPS-ASEC raw data. Consistent with Figures 4, and 5, the data show an increase in the probability of becoming newly uninsured and declines in out-of-pocket premium spending following the policy change. These results are similar across specifications and do not appear to be driven by pre-existing differences between treatment and control states. These patterns are broadly consistent with the regression estimates and provide additional evidence against differential pre-existing trends.

Next, Appendix Table C.1 reports estimates for two alternative insurance outcomes: any insurance coverage and currently uninsured. Column 1 shows that repealing the ACA individual mandate reduced the probability of having health insurance by 1.56 percentage points, equivalent to approximately a 2 percent decline relative to the pre-reform mean in states without a mandate. Consistent with this result, Column 2 shows that the repeal increased the probability of being uninsured by 1.56 percentage points, or approximately 9 percent relative to the pre-reform mean. These findings closely mirror the baseline estimates and reinforce the conclusion that repealing the individual mandate reduced insurance coverage.

Table C.2 reports estimates from the two-way fixed effects model, which does not explicitly account for the mass of zeros in out-of-pocket premium spending. The estimates are broadly consistent with the baseline results from the two-part model, with modest differences in magnitude. Similarly, the event-study estimates in Appendix Figure C.1 are nearly identical across the two approaches. Together, these results show that the main findings are robust to alternative approaches for modeling out-of-pocket premium spending.

Appendix Table C.3 and Figure C.2 report the effects of mandate repeal on medical out-of-pocket spending. The repeal reduced annual medical out-of-pocket spending by $346.99 in the full sample and by $396.09 among relatively healthy individuals, with no statistically significant effect among relatively sick individuals. The estimates are similar when the sample is restricted to insured individuals. The event-study estimates show no evidence of differential pre-trends, supporting the parallel-trends assumption. Together, these findings are consistent with the larger increase in uninsurance among relatively healthy individuals documented in the main analysis.

Given the decline in medical out-of-pocket expenditure after the repeal of the ACA individual mandate, I next examine whether healthcare utilization also changed. Appendix Table C.4 and

Figure C.3 report regression and event-study estimates for routine checkups. The repeal reduced the probability of receiving a routine checkup, with the decline concentrated among relatively healthy individuals. In contrast, I find no statistically significant change among relatively sick individuals. This pattern is consistent with the larger increase in uninsurance response among relatively healthy individuals documented in the main analysis.

Appendix Table D.1 reports placebo tests using self-employment and labor force participation, two outcomes that should not be directly affected by the repeal of the individual mandate. The estimated effects on both outcomes are small and statistically indistinguishable from zero. These findings provide additional evidence that the main results are unlikely to reflect broader differential changes between treatment and control states occurring around the time of the repeal.

Appendix Table D.2 reports falsification tests for two groups that should be less affected by the repeal of the individual mandate: adults aged 65 and older and low-income non-elderly adults. Adults aged 65 and older generally have access to Medicare and are therefore less exposed to changes in the individual mandate. Low-income adults with household incomes below 138% of the federal poverty line are generally eligible for Medicaid and were exempt from the mandate penalty during the study period. For adults aged 65 and older, the estimated effect on becoming newly uninsured is small in magnitude (0.12 percentage points) but statistically significant. The corresponding estimate for low-income adults is not statistically significant. These findings provide additional support for the interpretation that the main estimates reflect the repeal of the individual mandate rather than broader changes occurring in repeal states.

Appendix Table D.3 examines the sensitivity of the main results to alternative samples and specifications. First, I expand the baseline sample to include individuals with household incomes below 138% of the federal poverty line, testing whether the results are sensitive to the income restriction used in the main analysis. Second, because the treatment group is substantially larger and more geographically diverse than the control group of Massachusetts, New Jersey, and the District of Columbia, I follow Soni (2022) and restrict the sample to states in the New England and Mid-Atlantic Census Divisions. This restriction provides a more geographically comparable set of treatment and control states. Finally, I exclude 2018 to account for potential anticipatory responses following the passage of the repeal in late 2017. Across these alternative specifications, the estimates are similar in magnitude to the baseline results. The estimated effects generally retain their signs and remain statistically significant across the alternative specifications, although precision varies.

Table D.4 further examines whether the baseline estimates are driven by any single control state. I re-estimate the main specification after sequentially excluding Massachusetts, New Jersey, and the District of Columbia from the control group. The estimated effects on newly uninsured remain positive across all three specifications, while the effects on out-of-pocket premium spending remain negative and statistically significant. The magnitudes are also broadly similar to the baseline estimates. These results indicate that the main findings are not driven by any single control state.

A related concern is that the baseline specification compares a large and geographically heterogeneous group of treated states with only three control jurisdictions. As an additional sensitivity analysis, Appendix Table D.5 re-estimates the main specification using smaller subsets of treated states that differ in their traditional political orientation. [8] Columns 1 and 2 restrict the treatment group to California and New York, two populous traditionally Democratic-leaning states. Columns 3 and 4 restrict the treatment group to Texas and Florida, two populous traditionally Republican-leaning states, while Columns 5 and 6 use Georgia and Pennsylvania, two politically competitive states. In each specification, Massachusetts, New Jersey, and the District of Columbia remain the control jurisdictions.

The estimated effect on becoming newly uninsured remains positive across all three treatment-state samples and is statistically significant at the 5 percent level for California and New York and for Texas and Florida, and at the 10 percent level for Georgia and Pennsylvania. The estimated effect on out-of-pocket premium spending also remains negative and statistically significant across

[8] I consider political orientation because prior research documents partisan differences in attitudes toward and participation in the ACA. Trachtman (2019), for example, documents political polarization surrounding the ACA. The 2016 Kaiser Health Tracking Poll similarly reports substantially greater opposition to the individual mandate among Republicans than among Democrats.

all three samples. The estimates are consistent in sign with the baseline estimates, although their magnitudes vary across the alternative treatment-state samples. These results indicate that the baseline findings are not driven solely by the size or composition of the treatment group. However, this exercise does not eliminate the inference concern arising from having only three control jurisdictions; rather, it provides complementary evidence that the baseline results are not sensitive to the composition of the treatment group.

Appendix Figure D.1 plots U.S. Google search activity for ACA individual mandate" and Individual Mandate Repeal" from January 2012 through January 2025. Searches for "Individual Mandate Repeal" increased sharply in December 2017, coinciding with the passage of the repeal (Soni, 2022). Search activity declined thereafter but remained elevated relative to the pre-repeal period. The sharp increase in attention following passage of the repeal motivates the specification that excludes 2018 to account for potential anticipatory responses.

7 Concluding Remarks

A key justification for the Affordable Care Act (ACA) individual mandate penalty is that it mitigates adverse selection by encouraging both healthy and sick individuals to participate in the health insurance market. At the same time, critics argue that the mandate restricts individual choice and imposes financial burdens on individuals with relatively low demand for insurance. Evaluating these competing perspectives requires understanding not only how the mandate affects insurance participation, but also how its repeal affects coverage and out-of-pocket premium spending across consumers by health status.

To guide my empirical analysis, I build on the workhorse model of Geruso et al. (2023a), which distinguishes between insurance demand at the extensive margin (whether to purchase insurance) and the intensive margin (the generosity of insurance coverage). The model predicts that changes in mandate policy can affect both the decision to purchase insurance and the choice of insurance generosity, with implications for equilibrium premiums and welfare. My analysis applies this framework to the repeal of the ACA individual mandate in order to generate testable predictions about changes in insurance coverage and out-of-pocket premium payments driven by the repeal.

I test these predictions using data from the CPS ASEC and RWJF health exchange. I find evidence consistent with an increase in adverse selection following the repeal of the ACA individual mandate. The estimated coverage response is statistically significant among relatively healthy consumers but not among relatively sick consumers, although the difference between the subgroup estimates is not statistically significant. I further show that although Marketplace premiums increased following the repeal, average out-of-pocket premium spending declined, particularly among healthier consumers. Together, these results document divergent reduced-form changes in gross Marketplace benchmark premiums and reported consumer premium expenditures, suggesting that increases in gross Marketplace premiums do not necessarily translate into higher premium payments for consumers.

I then use the reduced-form estimates to examine how the financial consequences of the elimination of an enforceable individual mandate are distributed across consumers. The increase in gross Marketplace premiums indicates an increase in the unsubsidized price of Marketplace coverage. At the same time, average out-of-pocket premium spending declines, with the decline statistically significant among relatively healthy consumers but not among relatively sick consumers. The divergence between higher gross Marketplace premiums and lower consumer premium spending is consistent with the ACA subsidy structure limiting eligible consumers' exposure to higher benchmark premiums, although changes in insurance source, plan choice, and the composition of insured individuals may also contribute to this pattern. Taken together, the results highlight an important distinction between changes in gross Marketplace premiums and the premium expenditures observed among consumers.

More broadly, the results have implications for the design of health insurance policy beyond the ACA. The estimated coverage response is statistically significant among relatively healthy consumers but not among relatively sick consumers, although the difference between these estimates is not statistically significant. The results also highlight the importance of distinguishing between changes in gross insurance prices and consumer premium expenditures. In subsidized insurance markets, increases in gross premiums need not translate directly into equivalent increases in consumer premium payments, as subsidies may limit eligible consumers' exposure to higher premiums. Evaluating policies aimed at mitigating adverse selection should therefore consider both changes in market premiums and enrollment and the institutional features that determine consumers' effective premium payments. Future research should examine the longer-run consequences of eliminating an enforceable individual mandate for insurer participation, market competition, and consumer welfare.

Table 1: Descriptive Statistics

	Control States		Treated States		DiD [(4) - (3)]-
	Pre-2019 (1)	2019 (2)	Pre-2019 (3)	2019 (4)	[(2) - (1)] (5)
Insurance Outcomes					
Any Coverage Now (0/1)	0.871	0.906^{b}	0.832^{+++}	0.849***^^^	-0.018‡‡‡
Newly Insured (0/1)	0.017	0.009^{b}	0.023^{+++}	0.0075***	-0.008‡‡‡
Newly Uninsured (0/1)	0.014	0.012	0.021^{+++}	0.025***^^^	0.006‡‡‡
Premium Spending (2015 U.S. $)	1,740.83	$2,044.19^{a}$	$1,610.69^{+++}$	1,601.07***	-312.98‡‡‡
Marketplace Premium Outcomes					
27yrs old Individual (2015 U.S. $)	4,485.72	$5,107.66^{a}$	$3,717.30^{+++}$	5,402.78***	1,063.54‡‡‡
50yrs old Individual (2015 U.S. $)	6,483.24	$7,539.95^{a}$	$6,097.12^{+++}$	8,712.71***	1,558.88‡‡‡
30yrs old Couple (2015 U.S. $)	10,286.73	$11,927.73^{a}$	$8,523.65^{+++}$	13,128.78***	2,964.13‡‡‡
Individual Characteristics					
Age (years)	40.79	40.62	40.15^{++}	40.21**^^	0.23
Male (0/1)	0.48	0.47	0.48^{++}	0.48^^^	0.01
Married (0/1)	0.46	0.43	0.49^{+++}	0.49^^^	0.03
Household Size (number of persons)	3.20	3.26^{b}	3.19^{+++}	3.21^^^	-0.04
White non-Hispanic (0/1)	0.52	0.50	0.55^{+++}	0.53***^^^	0.00
Black, non-Hispanic (0/1)	0.16	0.18	0.15^{+++}	0.15**^^^	-0.02
Other race, non-Hispanic (0/1)	0.09	0.10	0.08	0.08	-0.01
Hispanic (0/1)	0.23	0.22	0.23^{+++}	0.24***^^^	0.02
Less Than High School (0/1)	0.10	0.09	0.12^{+++}	0.12^^^	0.01
High School (0/1)	0.35	0.39^{b}	0.35	0.35	-0.04
Some College (0/1)	0.30	0.25^{b}	0.33^{+++}	0.32***^^^	0.04
College and Advance Degrees (0/1)	0.25	0.27	0.20^{+++}	0.21**^^^	-0.01
Working but not Self Employed (0/1)	0.69	0.68	0.69	0.69	0.01
Self-Employed (0/1)	0.06	0.05	0.06^{+++}	0.06^^^	0.01
Not working (0/1)	0.25	0.27^{c}	0.25	0.25	-0.02
Sample Size	6,591	1,456	153,965	34,750	

Note: Sample includes adults aged 19 to 64 years old with household income between 138% and 400% of the federal poverty line (N=196,762).Table displays means adjusted by CPS ASEC sampling weights. Unless otherwise noted, all variables are constructed from the 2015–2019 CPS ASEC. Data on Marketplace premiums are obtained from the 2015–2019 Robert Wood Johnson Foundation (RWJF) Health Insurance Exchange dataset.
Test of significance (Sample means)
Control States, pre- vs post-repeal difference (two-tailed t-test): $^{a}p<0.01$, $^{b}p<0.05$, $^{c}p<0.1$
Pre-repeal treated vs. control states difference (two-tailed t-test): $^{+++}p<0.01$, $^{++}p<0.05$, $^{+}p<0.1$
Treated States, pre- vs post-repeal difference (two-tailed t-test): $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.1$
Treated vs Control States difference (two-tailed t-test): ^^^p<0.01, ^^p<0.05, ^p<0.1
Test of significance (DiD estimates): ‡‡‡p<0.01, ‡‡p<0.05, ‡p<0.1.

Table 2: Regression Results for Insurance Outcomes (Source: CPS ASEC, 2015 to 2019)

	Newly Uninsured			Newly Insured		
	All	Healthy	Sick	All	Healthy	Sick
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment X Post2019	0.0056***	0.0057***	0.0039	-0.0079***	-0.0087***	-0.0020
	(0.0018)	(0.0014)	(0.0075)	(0.010)	(0.0015)	(0.0071)
Sample Size	196,762	172,899	23,863	196,762	172,899	23,863
Pre-2019 Mean	0.0208	0.0213	0.0167	0.023	0.022	0.028
Year FE	✓	✓	✓	✓	✓	✓
State FE	✓	✓	✓	✓	✓	✓
Individual Controls	✓	✓	✓	✓	✓	✓

Notes: The table reports estimates of the effect of the ACA individual mandate repeal on the probability of becoming newly uninsured (Columns 1–3) and newly insured (Columns 4–6). Estimates are reported for the full sample and separately for relatively healthy and relatively sick individuals. Relatively healthy individuals are those reporting excellent, very good, or good health, while relatively sick individuals are those reporting fair or poor health. The sample includes adults aged 19 to 64 with household incomes between 138% and 400% of the federal poverty line. All regressions include individual controls, state fixed effects, and year fixed effects. Individual controls include age, sex, marital status, household size, race/ethnicity, educational attainment, and employment status. Estimates are weighted using CPS ASEC sampling weights. Standard errors are clustered at the state level and reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 3: Regression Estimate for Risk Composition (Source: CPS ASEC, 2015 to 2019)

	Sick Share	Healthy Share
	(1)	(2)
Treatment X Post2019	0.013	-0.013
	(0.010)	(0.010)
Sample Size	165,529	165,529
Pre-2019 Mean	0.12	0.88
Year FE	✓	✓
State FE	✓	✓
Individual Controls	✓	✓

Note: The table reports estimates of the effect of the ACA individual mandate repeal on the health-risk composition of the insured population, measured by the shares of insured individuals who are relatively healthy and relatively sick. The sample includes insured adults aged 19 to 64 with household incomes between 138% and 400% of the federal poverty line (N=165,529). All regressions control for age, sex, marital status, household size, race/ethnicity, educational attainment, employment status, state fixed effects, and year effects. Estimates are weighted using CPS ASEC sampling weights. Standard errors are clustered at the state level and reported in parentheses. ***p<0.01, **p<0.05, *p<0.10.

Table 4: Regression Estimates for Out-of-Pocket Premium Spending (Source: CPS ASEC, 2015 to 2019)

	Full Sample			Insured Only		
	All	Healthy	Sick	All	Healthy	Sick
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment X Post2019	-284.60***	-282.59***	-233.74	-282.76***	-286.18***	-186.92
	(35.93)	(43.05)	(200.43)	(35.08)	(41.292)	(227.07)
Sample Size	196,762	172,899	23,863	165,529	144,790	20,739
Pre-2019 Mean	1,615.97	1,660.90	1,299.28	1,831.48	1,896.06	1,398.50
Year FE	✓	✓	✓	✓	✓	✓
State FE	✓	✓	✓	✓	✓	✓
Individual Controls	✓	✓	✓	✓	✓	✓

Note: The table reports average marginal effects from two-part models of annual out-of-pocket premium spending. The first part estimates the probability of positive premium spending using a logit model, and the second part models spending conditional on positive spending using a gamma GLM with a log link. The two components are combined to obtain unconditional predicted premium spending, and the reported treatment effects are average marginal effects expressed in annual dollars. Columns 1–3 use the full sample, while Columns 4–6 restrict the sample to respondents with current health insurance coverage. Estimates are reported for the full health-status sample and separately for relatively healthy and relatively sick respondents. The baseline sample includes adults aged 19 to 64 with household incomes between 138% and 400% of the federal poverty line. All specifications include individual controls, state fixed effects, and year fixed effects and are weighted using CPS ASEC sampling weights. Standard errors are clustered at the state level and reported in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.10$.

Table 5: Regression Estimates for Market Premium (Source: RWJF Health Exchange 2015 to 2019)

	27yrs-old	50yrs-old	Coulple 30yrs-old
	(1)	(2)	(3)
Treatment X Post2019	+738.93***	+1,361.46***	+3,341.60***
	(110.98)	(210.99)	(426.12)
Sample Size	293,967	293,967	293,967
Pre-2019 Mean	3,478.95	5,819.43	11,851.59
Year FE	✓	✓	✓
State FE	✓	✓	✓

Notes: Notes: The table reports estimates of the effect of the ACA individual mandate repeal on annual benchmark Marketplace premiums. Estimates are reported for a 27-year-old, a 50-year-old, and a 30-year-old couple. All regressions include state and year fixed effects. Estimates are weighted using the Robert Wood Johnson Foundation (RWJF) Health Exchange Data Sampling weight. Standard errors are clustered at the state level and reported in parentheses. ***p<0.01, **p<0.05, *p<0.10.

Table 6: Heterogeneity Analysis for Healthy Consumers (Source: CPS ASEC, 2015 to 2019)

	Newly Uninsured (1)	Newly Insured (2)	Out of Pocket Premium Spending (3)	Sample Size (4)
Healthy Baseline Sample	0.0057 ***	-0.0087***	-282.59***	172,899
	(0.004)	(0.0015)	(43.05)	
By Sex				
Male	0.0059***	-0.0022	-177.26***	82,175
	(0.0019)	(0.0026)	(53.04)	
Female	0.0054**	-0.0141***	-357.99***	90,724
	(0.0023)	(0.0040)	(58.06)	
By Marital Status				
Married	0.0020	-0.001	-453.70***	93,156
	(0.0018)	(0.0015)	(65.31)	
Not Married	0.0093***	-0.0157***	-157.44***	79,743
	(0.0023)	(0.0036)	(44.02)	
By Parental Status				
Parents	0.0041	-0.0093***	-287.71***	98,393
	(0.0030)	(0.0016)	(90.68)	
Childless Adults	0.0073***	-0.0084***	-262.41***	74,506
	(0.0026)	(0.0020)	(51.38)	
By Employment Status				
Working but not Self-employed	0.0026	-0.0007	-410.12***	46,065
	(0.0048)	(0.0029)	(103.33)	
Self-employed	0.0067***	-0.0116***	-233.30***	126,834
	(0.0018)	(0.0013)	(51.21)	
Not Working	-	-	-	
By Age				
19-26	0.0191*	-0.0274**	-101.28	33,113
	(0.0104)	(0.0103)	(107.95)	
27-39	-0.0063	0.0052	-491.91***	59,190
	(0.0116)	(0.0053)	(109.52)	
40-49	0.0009	-0.0022	-486.67**	38,569
	(0.0041)	(0.0027)	(195.89)	
50-59	0.0063**	-0.0022	-50.69	29,421
	(0.0028)	(0.0029)	(93.29)	
60-64	0.0274***	-0.0318***	-231.97	12,606
	(0.0059)	(0.0043)	(403.62)	

Note: The table reports heterogeneity in the effects of the ACA individual mandate repeal on becoming newly uninsured, becoming newly insured, and out-of-pocket premium spending. Estimates are reported separately by sex, marital status, parental status, self-employment status, and age group. The sample includes adults aged 19 to 64 with household incomes between 138% and 400% of the federal poverty line. All regressions include individual controls, state fixed effects, and year effects. Estimates are weighted using CPS ASEC sampling weights. Standard errors are clustered at the state level and reported in parentheses. ***$p<0.01$, **$p<0.05$, *$p<0.10$.

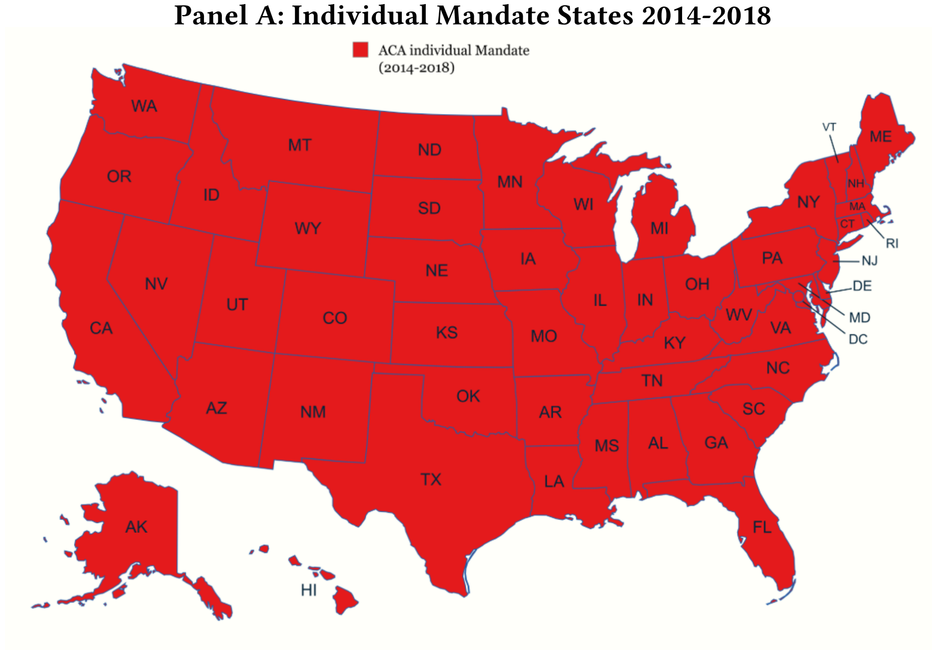
Table 7: Individual Mandate Repeal and Out-of-Pocket Premium Spending by Household Income (Source: CPS ASEC, 2015 to 2019)

	Baseline (1)	138%–150% FPL (2)	150%–200% FPL (3)	200%–250% FPL (4)	250%–300% FPL (5)	300%–400% FPL (6)
Treatment X Post2019	-284.60***	-161.76	-246.92**	-247.32**	120.69	-566.18***
	(35.93)	(198.62)	(122.01)	(117.87)	(133.29)	(116.02)
Sample Size	196,762	9,342	39,725	39,618	38,087	69,967
Pre-2019 Mean	1,615.97	982.30	1,159.16	1,476.82	1,673.77	2002.69
Year FE	✓	✓	✓	✓	✓	✓
State FE	✓	✓	✓	✓	✓	✓
Individual Controls	✓	✓	✓	✓	✓	✓

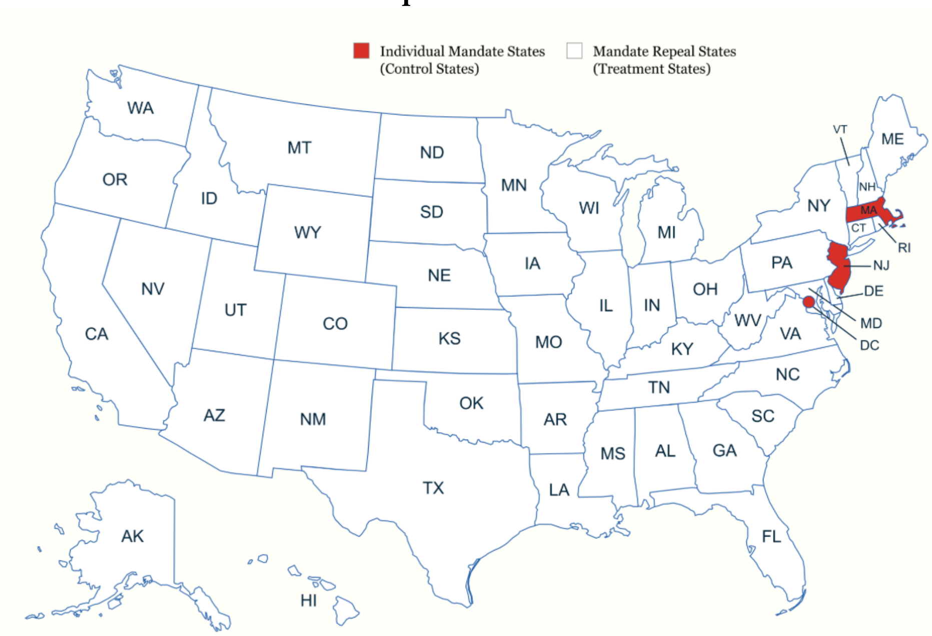
Note: The table reports heterogeneity in the effect of the ACA individual mandate repeal on out-of-pocket premium spending by household income. Column 1 reports the baseline estimate for individuals with household incomes between 138% and 400% of the federal poverty line, while Columns 2–6 report estimates separately by household income group. The sample includes adults aged 19 to 64. All regressions include individual controls, state fixed effects, and year effects. Estimates are weighted using CPS ASEC sampling weights. Standard errors are clustered at the state level and reported in parentheses. ***p<0.01, **p<0.05, *p<0.10.

Figure 1: Individual Mandates and Repealed States as of December 2019

Panel A: Individual Mandate States 2014-2018



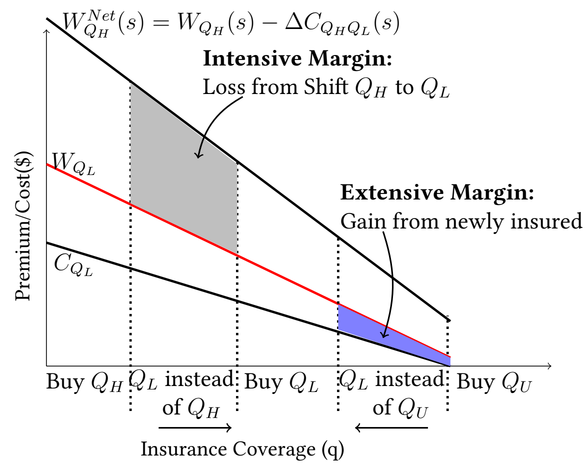
Panel B: Mandate Repeal States as of December 2019



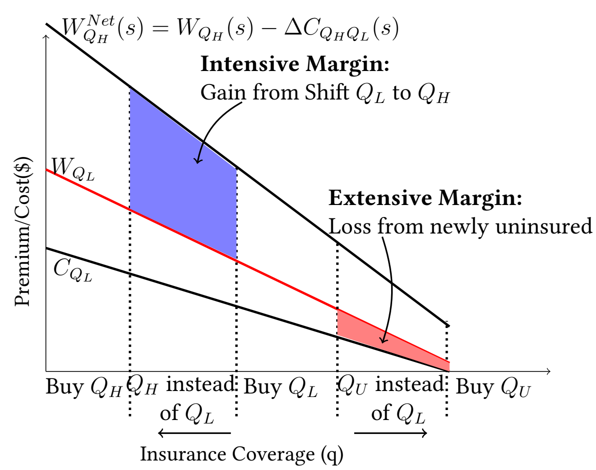
Note: Panel A of Figure 1 shows the implementation of the ACA individual mandate across the 50 states and the District of Columbia between 2014 and 2018. Panel B identifies the treatment states (those without an individual mandate in white) and the control states (those retaining an individual mandate in red) following the repeal of the federal mandate penalty. After the federal individual mandate penalty was reduced to zero in January 2019, Massachusetts retained its existing state mandate, while New Jersey and the District of Columbia implemented state-level individual mandates, leaving 48 states without mandates.

Figure 2: Effects of Adverse Selection on Consumer

Panel A: Consumer Surplus Under Mandate Penalty

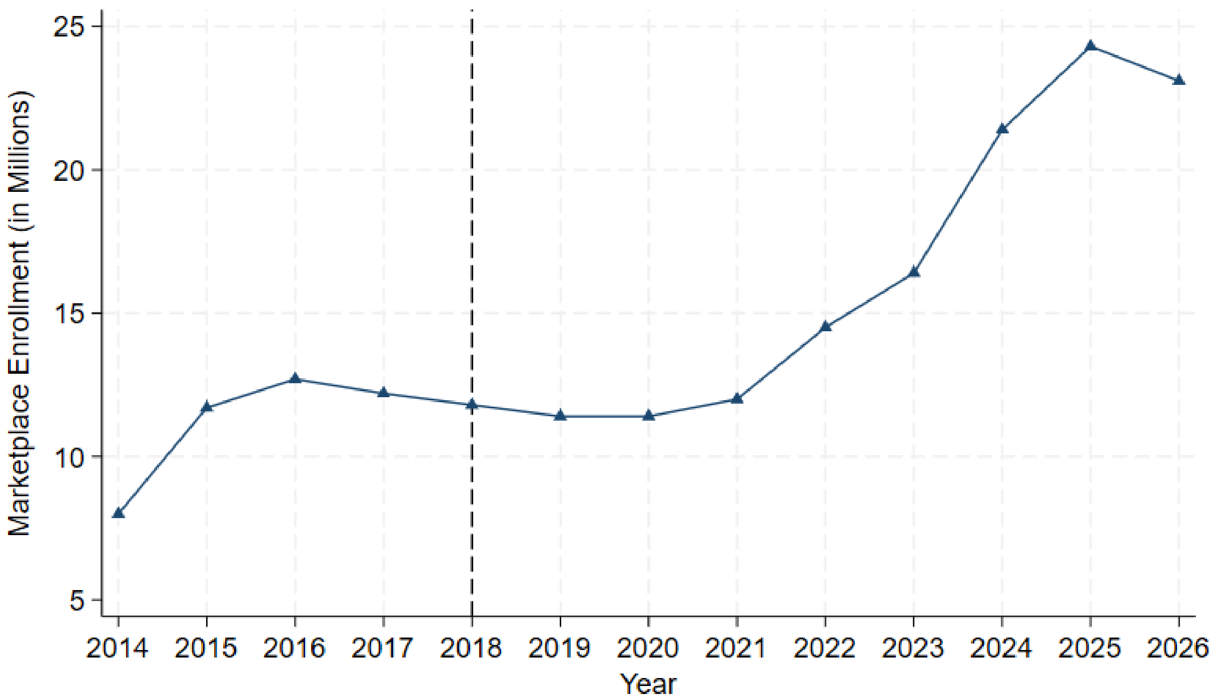


Panel B: Consumer Surplus Under Mandate Repeal



Note: Figure 2 illustrates the effects of the ACA individual mandate and its repeal on insurance choices and consumer surplus. Panel A shows that under the mandate penalty, relatively healthy consumers at the extensive margin move from being uninsured (Q_U) to the less generous plan (Q_L), while some relatively higher-risk consumers at the intensive margin move from the more generous plan (Q_H) to Q_L. Panel B shows that these responses operate in reverse following mandate repeal: relatively healthy consumers at the extensive margin move from Q_L to Q_U, while some relatively higher-risk consumers at the intensive margin move from Q_L to Q_H. The figure illustrates the corresponding gains and losses in consumer surplus associated with these extensive- and intensive-margin responses.

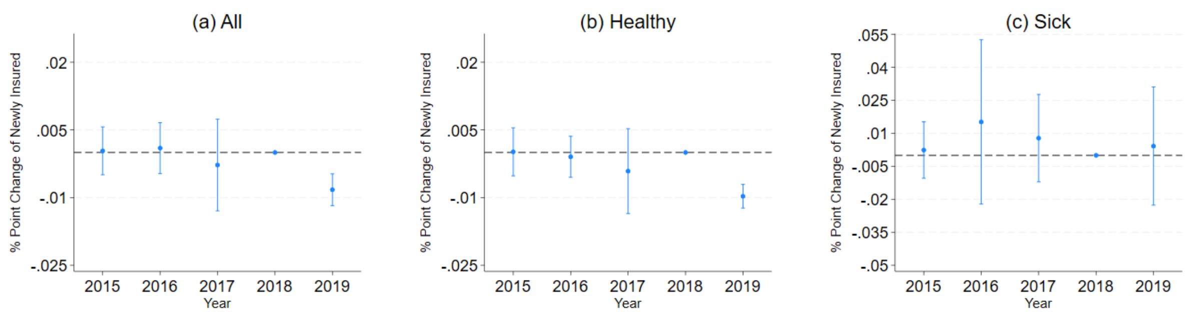
Figure 3: Trends in ACA Marketplace Enrollment



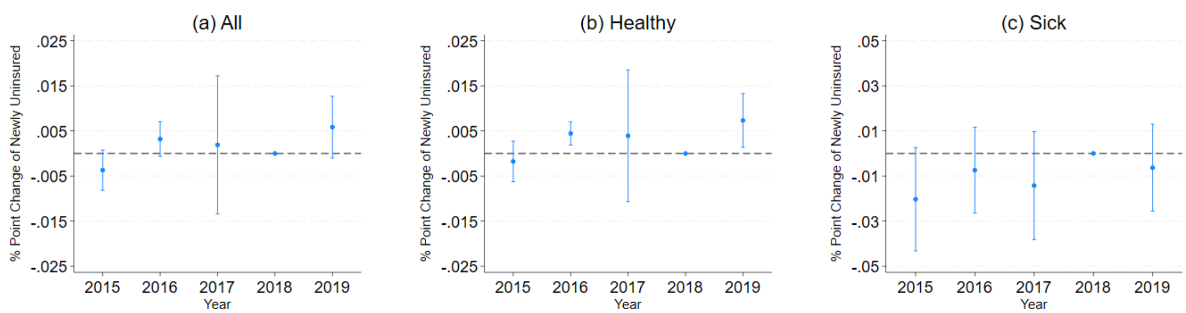
Note: Figure 3 shows trends in ACA Marketplace enrollment from 2015 to 2026 using annual open enrollment data from the Centers for Medicare and Medicaid Services (CMS). The solid line represents total Marketplace enrollment, and the vertical line marks the 2019 repeal of the federal individual mandate penalty. Marketplace enrollment includes enrollment through the federally facilitated Marketplace and state-based health insurance exchanges. (Source: Center for Medicare and Medicaid Services(CMS) Report, 2014 to 2026).

Figure 4: ACA Individual Mandate and Insurance Coverage by Health Status

Panel A: Event Study for Newly Insured



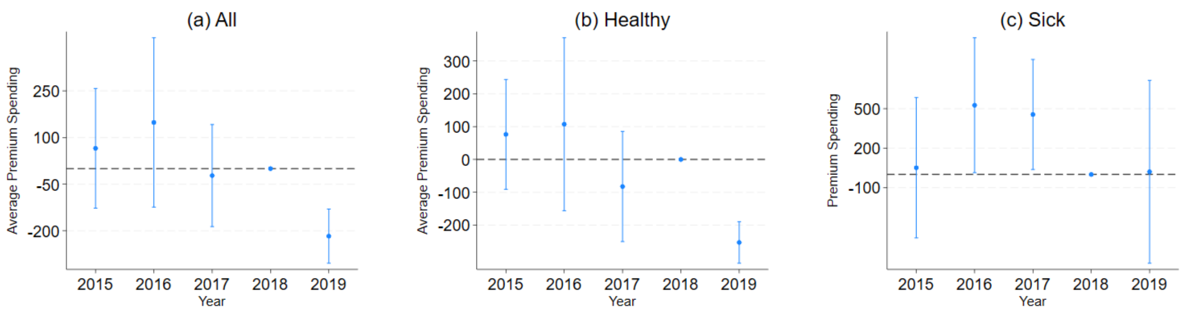
Panel B: Event Study for Newly Uninsured



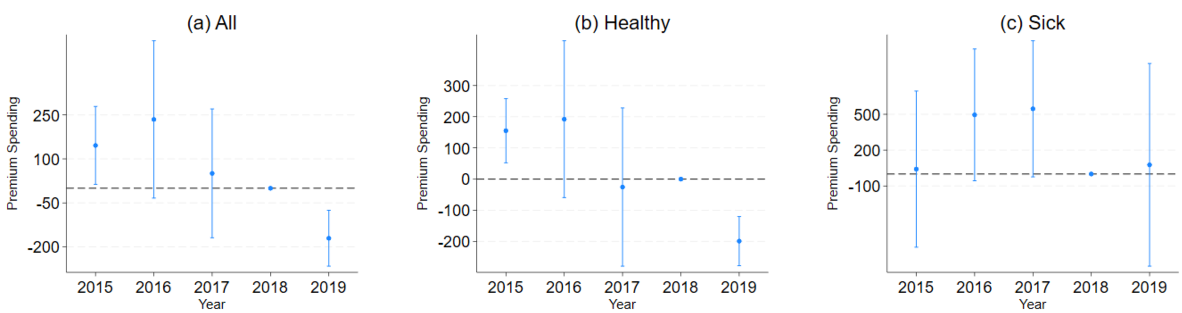
Note: Figure 4 presents event-study estimates of the effects of the ACA individual mandate repeal on insurance coverage, comparing states without individual mandates with states that retained mandates. Panel A reports estimates for the probability of becoming newly insured, while Panel B reports estimates for the probability of becoming newly uninsured. Estimates are reported for the full sample and separately for relatively healthy and sick individuals. The sample includes adults aged 19 to 64 with household incomes between 138% and 400% of the federal poverty line. Estimates are normalized to zero in 2018, the year immediately preceding the repeal. Points represent coefficient estimates, and vertical bars denote 95% confidence intervals. All regressions include individual controls, state fixed effects, and year effects. Estimates are weighted using CPS ASEC sampling weights. Standard errors are clustered at the state level.

Figure 5: ACA Individual Mandate Repeal and Out-of-Pocket Premium Spending by Health status

Panel A: Event Study for Full Sample



Panel B: Event Study for Insured Sample Only



Note: Figure 5 presents event-study estimates of the effects of the ACA individual mandate repeal on out-of-pocket premium spending, comparing states without individual mandates with states that retained mandates. Panel A reports estimates for the full sample, while Panel B restricts the sample to insured individuals. Within each panel, estimates are reported for the overall sample and separately for relatively healthy and sick individuals. The full sample includes adults aged 19 to 64 with household incomes between 138% and 400% of the federal poverty line. Estimates are normalized to zero in 2018, the year immediately preceding the repeal. Points represent coefficient estimates, and vertical bars denote 95% confidence intervals. All regressions include individual controls, state fixed effects, and year effects. Estimates are weighted using CPS ASEC sampling weights. Standard errors are clustered at the state level.

References

Akerlof, George A, "The market for "lemons": Quality uncertainty and the market mechanism," in "Uncertainty in economics," Elsevier, 1970, pp. 235–251.

Belotti, Federico, Partha Deb, Willard G Manning, and Edward C Norton, "twopm: Two-part models," *The Stata Journal*, 2015, *15* (1), 3–20.

Castaneda, Marco A and James Marton, "Employer-provided health insurance and the adverse selection problem," *Public Finance Review*, 2013, *41* (1), 3–36.

Courtemanche, Charles, James Marton, Benjamin Ukert, Aaron Yelowitz, and Daniela Zapata, "Early impacts of the Affordable Care Act on health insurance coverage in Medicaid expansion and non-expansion states," *Journal of Policy Analysis and Management*, 2017, *36* (1), 178–210.

Einav, Liran, Amy Finkelstein, and Mark R Cullen, "Estimating welfare in insurance markets using variation in prices," *The quarterly journal of economics*, 2010, *125* (3), 877–921.

_ and _ , "Selection in insurance markets: Theory and empirics in pictures," *Journal of Economic perspectives*, 2011, *25* (1), 115–138.

Fiedler, Matthew, "The ACA's Individual Mandate In Retrospect: What Did It Do, And Where Do We Go From Here? A review of recent research on the insurance coverage effects of the Affordable Care Act's individual mandate.," *Health Affairs*, 2020, *39* (3), 429–435.

Flood, Sarah, Miriam King, Renae Rodgers, Steven Ruggles, J Robert Warren, and Michael Westberry, "Integrated public use microdata series, current population survey: Version 6.0 [dataset]," *Minneapolis, MN: IPUMS*, 2023, *10*, D030.

Frean, Molly, Jonathan Gruber, and Benjamin D Sommers, "Premium subsidies, the mandate, and Medicaid expansion: Coverage effects of the Affordable Care Act," *Journal of health economics*, 2017, *53*, 72–86.

Fung, Vicki, Catherine Y Liang, Julie Shi, Veri Seo, Lindsay Overhage, William H Dow, Alan M Zaslavsky, Bruce Fireman, Stephen F Derose, Michael E Chernew et al., "Potential effects of eliminating the individual mandate penalty in California," *Health affairs*, 2019, *38* (1), 147–154.

Geruso, Michael and Timothy J Layton, "Selection in health insurance markets and its policy remedies," *Journal of Economic Perspectives*, 2017, *31* (4), 23–50.

__ , __ , Grace McCormack, and Mark Shepard, "The two-margin problem in insurance markets," *Review of Economics and Statistics*, 2023, *105* (2), 237–257.

__ , Timothy Layton, and Adam Leive, "The incidence of adverse selection: theory and evidence from health insurance choices," Technical Report, National Bureau of Economic Research 2023.

Gruber, Jonathan, "The impacts of the Affordable Care Act: how reasonable are the projections?," *National Tax Journal*, 2011, *64* (3), 893–908.

Hackmann, Martin B, Jonathan T Kolstad, and Amanda E Kowalski, "Adverse selection and an individual mandate: When theory meets practice," *American Economic Review*, 2015, *105* (3), 1030–1066.

Handel, Ben, Igal Hendel, and Michael D Whinston, "Equilibria in health exchanges: Adverse selection versus reclassification risk," *Econometrica*, 2015, *83* (4), 1261–1313.

Kreider, Amanda R, Timothy J Layton, Mark Shepard, and Jacob Wallace, "Adverse selection and network design under regulated plan prices: Evidence from Medicaid," *Journal of Health Economics*, 2024, *97*, 102901.

Lurie, Ithai Z, Daniel W Sacks, and Bradley Heim, "Does the individual mandate affect insurance coverage? Evidence from tax returns," *American Economic Journal: Economic Policy*, 2021, *13* (2), 378–407.

Manning, Willard G and John Mullahy, "Estimating log models: to transform or not to transform?," *Journal of health economics*, 2001, *20* (4), 461–494.

Marton, James, Aaron Yelowitz, and Jeffery C Talbert, "Medicaid program choice, inertia and adverse selection," *Journal of health economics*, 2017, *56*, 292–316.

McCullagh, Peter, *Generalized linear models*, Routledge, 2019.

Pascale, Joanne, Angela Fertig, and Kathleen Call, "Validation of two federal health insurance survey modules after Affordable Care Act implementation," *Journal of official statistics*, 2019, *35* (2), 409–460.

Peterson, Graeme, "Adverse Selection and Technological Change: Evidence from Medicare Part D," 2025.

Rothschild, Michael and Joseph Stiglitz, "Equilibrium in competitive insurance markets: An essay on the economics of imperfect information," in "Uncertainty in economics," Elsevier, 1976, pp. 257–280.

Saltzman, Evan, *Managing Adverse Selection in Health Insurance Markets: Evidence from the California and Washington ACA Exchanges*, University of Pennsylvania, 2018.

__ , "Demand for health insurance: Evidence from the California and Washington ACA exchanges," *Journal of Health Economics*, 2019, *63*, 197–222.

Shepard, Mark, "Hospital network competition and adverse selection: evidence from the Massachusetts health insurance exchange," *American Economic Review*, 2022, *112* (2), 578–615.

Soni, Aparna, "The impact of the repeal of the federal individual insurance mandate on uninsurance," *International Journal of Health Economics and Management*, 2022, *22* (4), 423–441.

Trachtman, Samuel, "Polarization, participation, and premiums: how political behavior helps explain where the aca works, and where it doesn't," *Journal of Health Politics, Policy and Law*, 2019, *44* (6), 855–884.

Wilson, Charles, "A model of insurance markets with incomplete information," *Journal of Economic theory*, 1977, *16* (2), 167–207.

Young, Jeffrey, "Kyl: Health bill a "stunning assault on liberty."," *The Hill*, 2009.

Zhang, Jessica, "The Affordable Care Act's Two-Legged Stool: The Effects of Eliminating the Federal Individual Mandate on Health Insurance Coverage." PhD dissertation, Harvard University 2021.

Appendix

A Theory of Adverse Selection and Individual Mandate

In this section, I build a simple model by incorporating the features of the ACA individual mandate and premium subsidy into a general model of adverse selection built by Rothschild and Stiglitz (1976). This model captures consumers' behavior by risk type when the insurance mandate is turned on or off at the extensive margin. This modeling framework is motivated by the ACA's individual mandate and premium subsidy features, which primarily influence the demand for health insurance on the extensive margin.

A.1 Set Up

In the health insurance market, consumers purchase insurance from insurers to protect themselves against uncertain medical expenditures. I focus on consumers' insurance decisions and the implications of these decisions for the risk composition of the insured population.

Consider a consumer with initial endowment Y. The consumer faces two possible health states. If the consumer remains healthy, the endowment remains Y. If the consumer becomes sick, the consumer incurs a medical loss $L < Y$.

Let π_i denote the probability of illness for consumer type $i \in \{H, S\}$, where H denotes a relatively healthy consumer and S denotes a relatively sick consumer. I assume that $0 < \pi_H < \pi_S < 1$. Without health insurance, consumers wealth in the healthy and sick states is $(Y, Y - L)$, respectively. The expected utility of an uninsured consumer is therefore

$$EU[\pi_i, 0, 0, 0] = \underbrace{(1 - \pi_i)U(Y)}_{\text{Healthy State}} + \underbrace{\pi_i U(Y - L)}_{\text{Sick State}} \tag{A.1}$$

I assume that the utility function is increasing and strictly concave, such that $U' > 0$ and $U'' < 0$. Consumers are therefore risk averse. Suppose that an insurance contract covers a fraction $\alpha \in [0, 1]$ of the medical loss L. The insurance benefit in the sick state is $q = \alpha L$, and the consumer is responsible for the remaining medical expenses $(1 - \alpha)L$.

Let p denote the total premium paid for the insurance contract. With insurance, consumer's wealth in the healthy and sick states is therefore $Y - p$ and $Y - p - (1 - \alpha)L$, respectively. The expected utility from purchasing insurance is

$$EU[\pi_i, q, 0, 0] = \underbrace{(1 - \pi_i)U(Y - p)}_{\text{Healthy State}} + \underbrace{\pi_i U\left[Y - p - (1 - \alpha)L\right]}_{\text{Sick State}} \tag{A.2}$$

In the absence of an individual mandate, consumers may choose whether to purchase insurance or not. Consumer type i will choose to purchase insurance when the expected utility from being insured is at least better than the expected for not having insurance given by:

$$EU[\pi_i, q, 0, 0] \geq EU[\pi_i, 0, 0, 0]. \tag{A.3}$$

Risk aversion alone does not imply that all consumers purchase insurance. The decision to buy insurance also depends on the premium, the generosity of the coverage, and the consumer's probability of illness. However, relatively healthy consumers may choose to remain uninsured when the utility gain from insurance is insufficient to offset the premium required for coverage.

A.1.1 Government Intervention through Individual Mandate and Premium Subsidy

I next introduce the individual mandate penalty and the premium subsidy provisions of the Affordable Care Act (ACA). The mandate increases the cost of remaining uninsured, while premium subsidies reduce the amount that eligible consumers pay for insurance. These policy instruments can affect insurance participation and, consequently, the risk composition of the insured population.

Because the ACA prohibits insurers from setting premiums based on consumers' health status, I assume that healthy and sick consumers purchasing the same insurance contract face a common pooled premium. Let θ denote the share of relatively sick consumers among the insured population. The average illness probability among insured consumers is:

$$\bar{\pi}(\theta) = (1 - \theta)\pi_H + \theta\pi_S. \tag{A.4}$$

Under competitive pricing and abstracting from administrative costs, insurers set the pooled premium equal to expected insurance payments. Because the insurance contract covers a fraction α of the medical loss, the break-even premium is:

$$p(\theta) = \bar{\pi}(\theta)\alpha L. \tag{A.5}$$

Since $\pi_S > \pi_H$, the pooled premium increases with the share of relatively sick consumers in the insured population:

$$\frac{\partial p(\theta)}{\partial \theta} = \alpha L(\pi_S - \pi_H) > 0. \tag{A.6}$$

This relationship should be interpreted as a conditional comparative static. The framework does not solve endogenously for the equilibrium value of θ. Rather, it illustrates how a change in the health composition of the insured population affects the pooled premium. Conditional on relatively healthy consumers entering the insurance market, θ falls and the average expected medical cost of the insured pool declines, placing downward pressure on the pooled premium. Conversely, conditional on relatively healthy consumers exiting the market, θ rises and the pooled premium increases.

I next incorporate the individual mandate and premium subsidy. Let τ denote the mandate penalty imposed on an uninsured consumer. For an eligible insured consumer, let the premium subsidy be $s(p(\theta), Y_i)$, where the subsidy may depend on the benchmark premium and the consumer's income. The consumer therefore pays the net premium

$$p_i^{net}(\theta) = p(\theta) - s(p(\theta), Y_i)\,. \tag{A.7}$$

For consumers eligible for premium tax credits, an increase in the benchmark premium may increase the subsidy. Consequently, the change in the consumer's net premium associated with a change in the gross premium is:

$$\frac{\partial p_i^{net}(\theta)}{\partial p(\theta)} = 1 - \frac{\partial s(p(\theta), Y_i)}{\partial p(\theta)}. \tag{A.8}$$

When $\frac{\partial s(p(\theta),Y_i)}{\partial p(\theta)} > 0$, the consumer's net premium increases by less than the corresponding increase in the gross premium. Thus, the subsidy can attenuate, although it need not eliminate, the pass-through of higher pooled premiums to eligible consumers. For consumers who are not eligible for a subsidy, $s = 0$, so changes in the gross premium are fully reflected in the premium they face.

Under the individual mandate and premium subsidy, the expected utility from purchasing insurance is:

$$EU[\pi_i, q, s, 0] = \underbrace{(1-\pi_i)U\left[Y-(p(\theta)-s(p(\theta),Y_i))\right]}_{\text{Healthy State}} + \underbrace{\pi_i U\left[Y-(p(\theta)-s(p(\theta),Y_i))-(1-\alpha)L\right]}_{\text{Sick State}}, \tag{A.9}$$

while the expected utility from remaining uninsured is:

$$EU[\pi_i, 0, 0, \tau] = \underbrace{(1-\pi_i)U(Y-\tau)}_{\text{Healthy State}} + \underbrace{\pi_i U(Y-\tau-L)}_{\text{Sick State}}, \tag{A.10}$$

where $i \in \{H, S\}$. The mandate penalty increases the relative cost of remaining uninsured, while the premium subsidy lowers the net premium faced by eligible insured consumers. Consumer i purchases insurance under the mandate when:

$$EU[\pi_i, q, s, 0] \geq EU[\pi_i, 0, 0, \tau]. \tag{A.11}$$

Risk aversion alone therefore does not imply universal enrollment under the mandate. Insurance participation depends on the consumer's health risk, the pooled premium, subsidy eligibility, mandate penalty, and the generosity of the insurance plan. For consumers at the margin of insurance participation, a sufficiently large mandate penalty or premium subsidy can make purchasing insurance more desirable than remaining uninsured.

A.1.2 Government Intervention through Mandate Repeal and Premium Subsidy

I next consider a setting in which the government eliminates the individual mandate penalty but continues to provide premium subsidies. Consumers can therefore choose between purchasing health insurance and remaining uninsured without facing a mandate penalty. Because ACA-compliant individual-market premiums cannot vary with consumers' health status, I abstract from other permitted rating factors and assume that healthy and sick consumers purchasing the same contract face a common pooled premium that reflects the expected medical costs of the insured population.

Consumers differ in their underlying health risk and fall into two categories: healthy and sick. Healthy consumers face a relatively low probability of illness, π_H, while sick consumers face a higher probability of illness, such that $\pi_S > \pi_H$. Let θ denote the share of sick consumers among the insured population. The average illness risk in the insured pool is therefore

$$\bar{\pi}(\theta) = (1-\theta)\pi_H + \theta\pi_S. \tag{A.12}$$

Assuming competitive insurers and abstracting from administrative costs, the pooled premium satisfies the insurer's break-even condition:

$$p(\theta) = \bar{\pi}(\theta)\alpha L, \tag{A.13}$$

where L denotes the medical loss when sick and $\alpha \in [0,1]$ denotes the share of the medical loss covered by insurance. Because $\pi_S > \pi_H$, an increase in the share of sick consumers among the insured raises average expected medical costs and therefore the pooled premium:

$$\frac{\partial p(\theta)}{\partial \theta} = \alpha L(\pi_S - \pi_H) > 0. \tag{A.14}$$

As in the preceding subsection, this relationship should be interpreted as a conditional comparative static rather than as a complete characterization of the equilibrium value of θ. Conditional on relatively healthy consumers exiting the insured pool, θ increases and the pooled premium rises.

Eligible consumers who purchase insurance pay the net premium

$$p(\theta) - s\left(p(\theta), Y_i\right), \tag{A.15}$$

where $s\left(p(\theta), Y_i\right)$ denotes the premium subsidy, which may depend on the benchmark premium and the consumer's income. The insurer continues to receive the gross premium $p(\theta)$, while the subsidy reduces the amount paid directly by the consumer.

Healthy Enrollees: To illustrate how eliminating the mandate penalty can affect insurance participation, consider relatively healthy consumers near the margin of enrollment. Let their expected utility from purchasing insurance be

$$
\begin{aligned}
I_H &\equiv EU_H[\pi_H, q, s, 0] \\
&= \underbrace{(1-\pi_H)U\left[Y-(p(\theta)-s(p(\theta),Y_H))\right]}_{\text{Healthy State}} + \underbrace{\pi_H U\left[Y-(p(\theta)-s(p(\theta),Y_H))-(1-\alpha)L\right]}_{\text{Sick State}}. \quad \text{(A.16)}
\end{aligned}
$$

Before repeal, a healthy consumer who remains uninsured pays the mandate penalty τ. Expected utility from remaining uninsured under the mandate is therefore

$$U_H^\tau \equiv EU_H[\pi_H, 0, 0, \tau] = (1-\pi_H)U(Y-\tau) + \pi_H U(Y-\tau-L). \quad \text{(A.17)}$$

After repeal, the mandate penalty is eliminated, so expected utility from remaining uninsured becomes

$$U_H^0 \equiv EU_H[\pi_H, 0, 0, 0] = (1-\pi_H)U(Y) + \pi_H U(Y-L). \quad \text{(A.18)}$$

For marginal relatively healthy consumers whose enrollment is induced by the mandate, I assume the following ordering:

$$U_H^0 > I_H \geq U_H^\tau. \quad \text{(A.19)}$$

The second inequality, $I_H \geq U_H^\tau$, implies that before repeal these consumers weakly prefer purchasing insurance to remaining uninsured and paying the mandate penalty. The first inequality, $U_H^0 > I_H$, implies that once the mandate penalty is eliminated, the same consumers strictly prefer remaining uninsured. The repeal therefore induces these marginal healthy consumers to switch from insurance to uninsurance.

Conditional on this exit of relatively healthy consumers, the share of sick consumers among the remaining insured population, θ, increases. The average expected medical cost of the insured pool consequently rises, and the insurer's break-even condition implies an increase in the pooled premium $p(\theta)$. This provides a stylized adverse-selection mechanism through which eliminating the mandate penalty can increase the risk of the insured pool and place upward pressure on gross insurance premiums.

Sick Enrollees: Next I consider relatively sick consumers, who face a higher probability of illness than relatively healthy consumers, such that $\pi_S > \pi_H$. Let their expected utility from purchasing insurance after mandate repeal be:

$$I_S \equiv EU_S[\pi_S, q, s, 0] = (1-\pi_S)U\left[Y - (p(\theta) - s(p(\theta), Y_S))\right] \\ + \pi_S U\left[Y - (p(\theta) - s(p(\theta), Y_S)) - (1-\alpha)L\right]. \tag{A.20}$$

Before the mandate repeal, a sick consumer who remains uninsured pays the mandate penalty τ. The expected utility from remaining uninsured under the mandate is therefore:

$$U_S^\tau \equiv EU_S[\pi_S, 0, 0, \tau] = (1-\pi_S)U(Y-\tau) + \pi_S U(Y-\tau-L). \tag{A.21}$$

After repeal, the mandate penalty is eliminated, so expected utility from remaining uninsured becomes

$$U_S^0 \equiv EU_S[\pi_S, 0, 0, 0] = (1-\pi_S)U(Y) + \pi_S U(Y-L). \tag{A.22}$$

For relatively sick consumers who continue to prefer insurance even in the absence of the mandate penalty, I assume the following ordering:

$$I_S > U_S^0 > U_S^\tau. \tag{A.23}$$

The inequality $U_S^0 > U_S^\tau$ follows because eliminating the mandate penalty increases the utility of remaining uninsured. However, unlike the marginal relatively healthy consumers described above, relatively sick consumers continue to prefer insurance after repeal because $I_S > U_S^0$. Since $\pi_S > \pi_H$, relatively sick consumers face greater expected medical losses and therefore have a greater expected benefit from insurance. Thus, although repeal makes remaining uninsured more attractive, it need not induce relatively sick consumers to exit coverage.

Taken together, the two cases illustrate the adverse-selection mechanism generated by mandate repeal.

For marginal relatively healthy consumers, $U_H^0 > I_H \geq U_H^\tau$, so eliminating the mandate penalty can induce a switch from insurance to uninsurance. In contrast, for relatively sick consumers satisfying $I_S > U_S^0 > U_S^\tau$, insurance remains the preferred option after repeal. Conditional on the exit of relatively healthy consumers, the share of sick consumers in the insured population increases, raising average expected medical costs and placing upward pressure on the pooled premium.

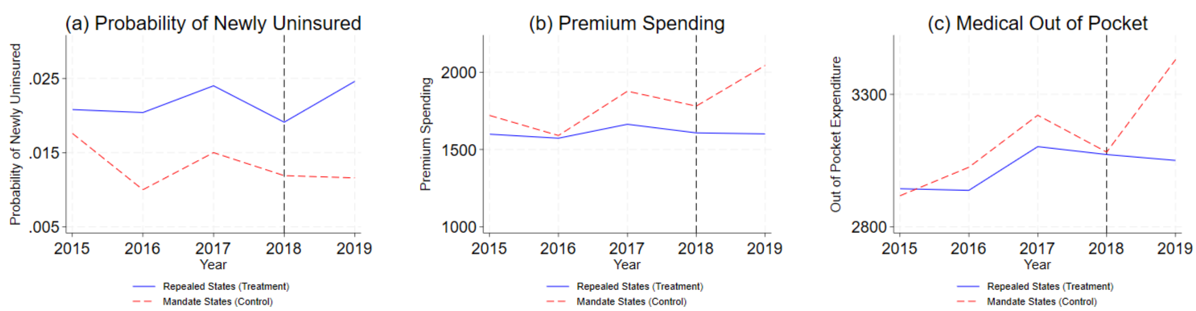
B Individual Mandates and Repeal States

Table B.1: States with Insurance Mandate and States without Mandate as of 2024

	2019	2020
Treatment	48 States	46 states
Control	MA, NJ, DC	MA, NJ, DC, CA, RI

Notes: The table summarizes the classification of states by individual mandate status in 2019 and 2020. Treatment states are states without an individual mandate in effect, while control states are jurisdictions with an individual mandate in effect. In 2019, Massachusetts, New Jersey, and the District of Columbia had individual mandates in effect, leaving 48 states in the treatment group. In 2020, California and Rhode Island also had state-level individual mandates in effect, reducing the treatment group to 46 states.

Figure B.1: Trends in Newly Uninsured and Healthcare Spending Outcomes



Note: Figure B.1 presents trends in insurance coverage and healthcare spending outcomes for the treatment and control states. Panel (a) shows trends in the probability of becoming newly uninsured, Panel (b) shows trends in out-of-pocket premium spending, and Panel (c) shows trends in medical out-of-pocket spending. Treatment states are states without an individual mandate, while control states are states with an individual mandate. The sample includes adults aged 19 to 64 with household incomes between 138% and 400% of the federal poverty line (N=196,672). Estimates are weighted using CPS ASEC sampling weights. Source: Current Population Survey Annual Social and Economic Supplement (CPS ASEC), 2015–2019.

C Main Results

Table C.1: Alternate Insurance Outcomes (Source: Current Population Survey ASEC, 2015 to 2019)

	Any Coverage Now (1)	Currently Uninsured (2)
Treatment X Post2019	-0.0156***	0.0156***
	(0.0046)	(0.0046)
Sample Size	196,762	196,762
Pre 2019 Mean	0.83	0.17
Year FE	✓	✓
State FE	✓	✓
Individual Controls	✓	✓

Note: The table reports estimates of the effect of the ACA individual mandate repeal on alternative measures of insurance coverage. Column 1 reports estimates for the probability of having any health insurance coverage, while Column 2 reports estimates for the probability of being currently uninsured. The sample includes adults aged 19 to 64 with household incomes between 138% and 400% of the federal poverty line. All regressions include individual controls, state fixed effects, and year effects. Estimates are weighted using CPS ASEC sampling weights. Standard errors are clustered at the state level and reported in parentheses. ***p<0.01, **p<0.05, *p<0.10.

Table C.2: Out-of-Pocket Premium Spending Using TWFE (Source: CPS ASEC, 2015 to 2019)

	Full Sample			Insured Only		
	All	Healthy	Sick	All	Healthy	Sick
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment X Post2019	-326.05***	-338.36***	-176.32	-319.20***	-335.63***	-120.74
	(33.65)	(41.56)	(243.38)	(33.62)	(53.38)	(244.97)
Sample Size	196,762	172,899	23,863	165,529	144,790	20,739
Pre-2019 Mean	1,615.97	1,660.90	1,299.28	1,831.48	1,896.06	1,398.50
Year FE	✓	✓	✓	✓	✓	✓
State FE	✓	✓	✓	✓	✓	✓
Individual Controls	✓	✓	✓	✓	✓	✓

Note: The table reports estimates of the effect of the ACA individual mandate repeal on out-of-pocket premium spending using a two-way fixed effects (TWFE) specification. Columns 1–3 report estimates for the full sample, while Columns 4–6 restrict the sample to insured individuals. Within each sample, estimates are reported for all individuals and separately for relatively healthy and sick individuals. The full sample includes adults aged 19 to 64 with household incomes between 138% and 400% of the federal poverty line. All regressions include individual controls, state fixed effects, and year effects. Estimates are weighted using CPS ASEC sampling weights. Standard errors are clustered at the state level and reported in parentheses. ***p<0.01, **p<0.05, *p<0.10.

Table C.3: Regression Estimates for Medical Out-of-Pocket Costs (Source: CPS ASEC, 2015 to 2019)

	Full Sample			Insured Only		
	All	Healthy	Sick	All	Healthy	Sick
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment X Post2019	-346.99***	-396.09***	63.24	-337.96***	-385.39***	70.87
	(49.61)	(60.73)	(528.34)	(46.63)	(78.27)	(531.30)
Sample Size	196,762	172,762	23,863	165,529	144,790	20,739
Pre-2019 Mean	3,014.73	2,997.81	3,133.98	3,303.40	3,192.36	3,283.95
Year FE	✓	✓	✓	✓	✓	✓
State FE	✓	✓	✓	✓	✓	✓
Individuals Control	✓	✓	✓	✓	✓	✓

Note: The table reports estimates of the effect of the ACA individual mandate repeal on medical out-of-pocket spending. Columns 1–3 report estimates for the full sample, while Columns 4–6 restrict the sample to insured individuals. Within each sample, estimates are reported for all individuals and separately for relatively healthy and sick individuals. The full sample includes adults aged 19 to 64 with household incomes between 138% and 400% of the federal poverty line ($N = 196{,}762$). All regressions include individual controls, state fixed effects, and year effects. Estimates are weighted using CPS ASEC sampling weights. Standard errors are clustered at the state level and reported in parentheses. ***p<0.01, **p<0.05, *p<0.10.

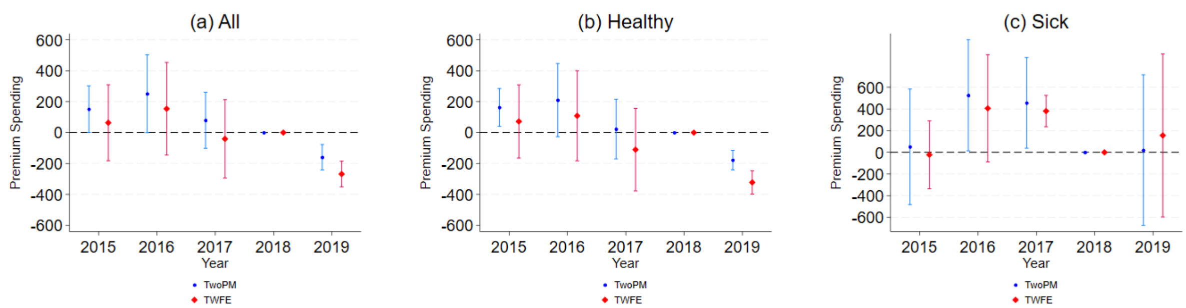
Table C.4: Regression Results for Routine Check Up (Source: BRFSS 2014 to 2019)

	All (1)	Healthy (2)	Sick (3)
Treatment X Post2019	-0.027**	-0.030**	-0.002
	(0.011)	(0.014)	(0.045)
Sample Size	190,208	154,689	35,127
Pre-2019 Mean	0.71	0.70	0.78
Year FE	✓	✓	✓
State FE	✓	✓	✓
Individual Controls	✓	✓	✓

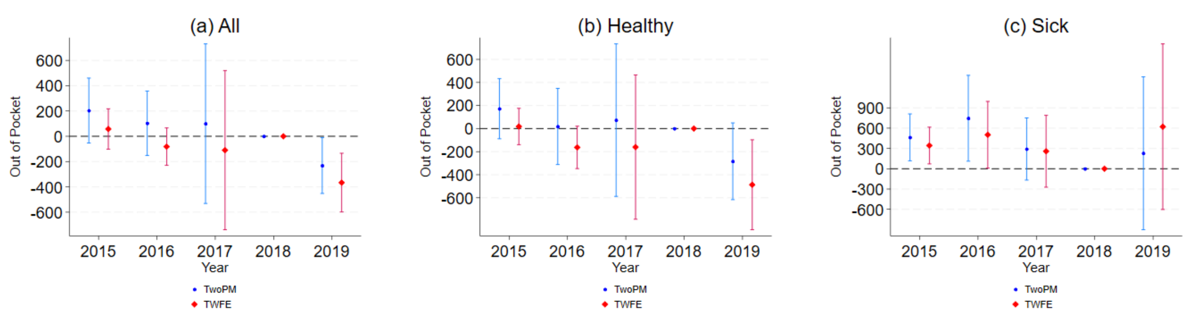
Note: The table reports estimates of the effect of the ACA individual mandate repeal on the probability of receiving a routine checkup. Estimates are reported for the full sample and separately for relatively healthy and sick individuals. The sample includes adults aged 19 to 64 with household incomes between 138% and 400% of the federal poverty line. All regressions include individual controls, state fixed effects, and year effects. Estimates are weighted using BRFSS sampling weights. Standard errors are clustered at the state level and reported in parentheses. ***p<0.01, **p<0.05, *p<0.10.

Figure C.1: Event Study for Out-of-Pocket Premium Spending

Panel A: Full Sample



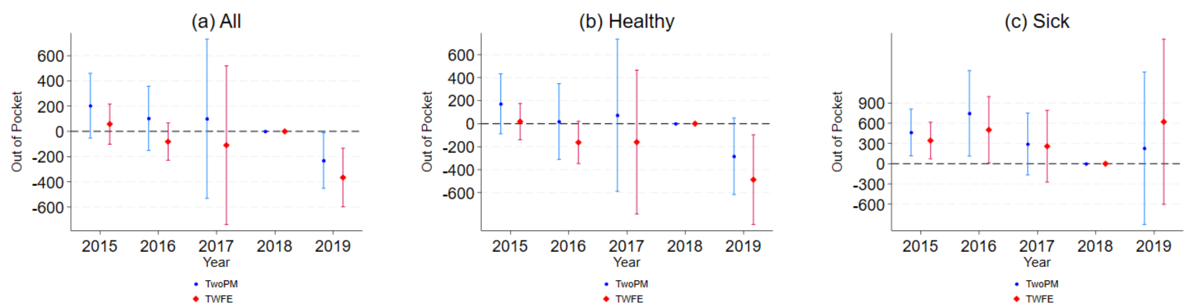
Panel B: Insured Sample Only



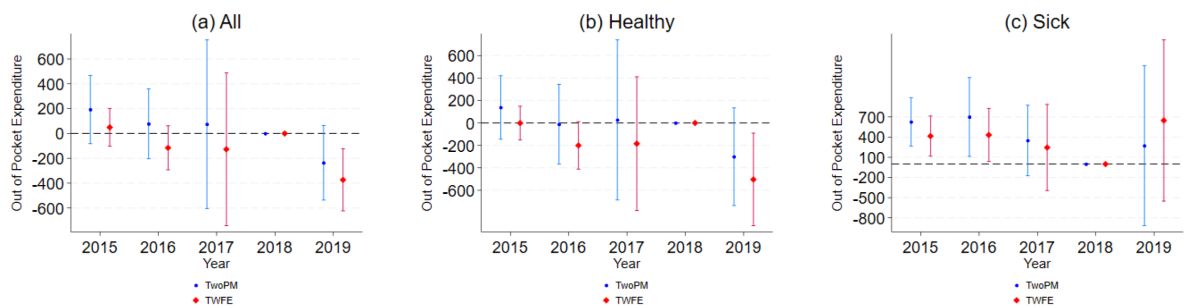
Note: Figure C.1 presents event-study estimates of the effect of the ACA individual mandate repeal on out-of-pocket premium spending. Panel A reports estimates for the full sample, while Panel B restricts the sample to insured individuals. The sample includes adults aged 19 to 64 with household incomes between 138% and 400% of the federal poverty line (N=196,762). Estimates are normalized to zero in 2018, the year immediately preceding the repeal. Points represent coefficient estimates, and vertical bars denote 95% confidence intervals. All regressions include individual controls, state fixed effects, and year effects. Estimates are weighted using CPS ASEC sampling weights. Standard errors are clustered at the state level. Source: Current Population Survey Annual Social and Economic Supplement (CPS ASEC), 2015–2019.

Figure C.2: Event Study for Medical Out of Pocket Expenditure

Panel A: Full Sample

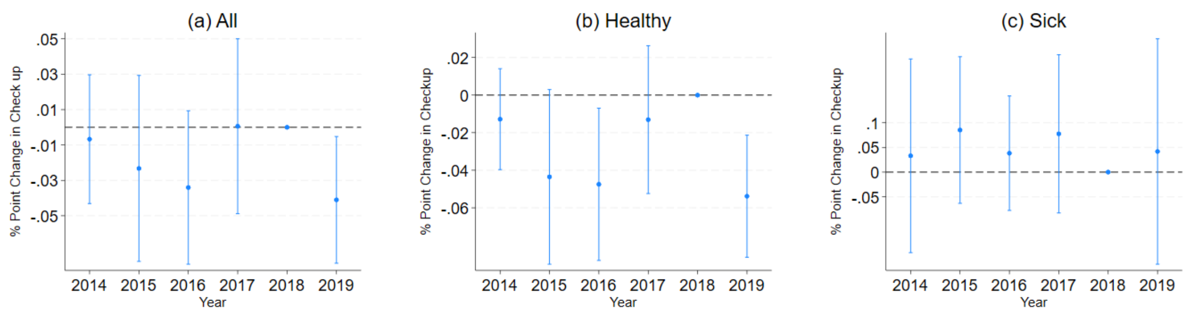


Panel B: Insured Sample Only



Note: Figure C.2 presents event-study estimates of the effect of the ACA individual mandate repeal on medical out-of-pocket spending. Panel A reports estimates for the full sample, while Panel B restricts the sample to insured individuals. The sample includes adults aged 19 to 64 with household incomes between 138% and 400% of the federal poverty line (N=189,850). Estimates are normalized to zero in 2018, the year immediately preceding the repeal. Points represent coefficient estimates, and vertical bars denote 95% confidence intervals. All regressions include individual controls, state fixed effects, and year effects. Estimates are weighted using CPS ASEC sampling weights. Standard errors are clustered at the state level. Source: Current Population Survey Annual Social and Economic Supplement (CPS ASEC), 2015–2019.

Figure C.3: Event Study for Routine Check up



Note: Figure C.3 presents event-study estimates of the effect of the ACA individual mandate repeal on the probability of receiving a routine checkup. Panel (a) reports estimates for the full sample, Panel (b) for relatively healthy individuals, and Panel (c) for relatively sick individuals. The sample includes adults aged 19 to 64 with household incomes between 138% and 400% of the federal poverty line (N=190,708). Estimates are normalized to zero in 2018, the year immediately preceding the repeal. Points represent coefficient estimates, and vertical bars denote 95% confidence intervals. All regressions include individual controls, state fixed effects, and year effects. Estimates are weighted using BRFSS sampling weights. Standard errors are clustered at the state level. Source: Behavioral Risk Factor Surveillance System (BRFSS), 2014–2019.

D Robustness Checks

Table D.1: Placebo Test for Alternate Outcomes (Source: CPS ASEC, 2015 to 2019)

	Self-employed (1)	In Labor Force (2)
Treatment X Post2019	3.36×10^{-17}	0.0005
	(2.28×10^{-17})	(0.0012)
Sample Size	149,195	195,250
Year FE	✓	✓
State FE	✓	✓
Individual Controls	✓	✓

Note: The table reports placebo estimates of the effect of the ACA individual mandate repeal on self-employment and labor force participation. Column 1 reports estimates for the probability of being self-employed, while Column 2 reports estimates for the probability of being in the labor force. The sample includes adults aged 19 to 64 with household incomes between 138% and 400% of the federal poverty line. All regressions include individual controls, state fixed effects, and year effects. Estimates are weighted using CPS ASEC sampling weights. Standard errors are clustered at the state level and reported in parentheses. ***p<0.01, **p<0.05, *p<0.10.

Table D.2: Falsification Test (Source: Current Population Survey ASEC, 2015 to 2019)

	Newly Uninsured		Premium Spending		Out of Pocket Expenditure	
	Elderly (Age >64)	Low-Income (<138% FPL)	Elderly (Age >64)	Low-Income (<138% FPL)	Elderly (Age >64)	Low-Income (<138% FPL)
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment X Post2019	-0.0012***	0.0054	56.46	-87.89	-103.90	-29.39
	(0.0003)	(0.0046)	(74.14)	(158.01)	(66.00)	(215.59)
Sample Size	54,962	76,937	54,962	76,937	54,962	76,937
Year FE	✓	✓	✓	✓	✓	✓
State FE	✓	✓	✓	✓	✓	✓
Individual Controls	✓	✓	✓	✓	✓	✓

Note: The table reports falsification estimates of the effect of the ACA individual mandate repeal on becoming newly uninsured, out-of-pocket premium spending, and medical out-of-pocket spending. Estimates are reported separately for adults aged 65 and older and low-income adults with household incomes below 138% of the federal poverty line. All regressions include individual controls, state fixed effects, and year effects. Estimates are weighted using CPS ASEC sampling weights. Standard errors are clustered at the state level and reported in parentheses. ***p<0.01, **p<0.05, *p<0.10.

Table D.3: Sensitivity Analyses (Source: Current Population Survey ASEC, 2015 to 2019)

	Newly Uninsured				Premium Spending				Medical Out of Pocket Expenditure			
	Baseline	Include Income 0–138% FPL Sample	New England & Mid-Altantic Divisions Only	Omit 2018	Baseline	Include Income 0–138% FPL Sample	New England & Mid-Altantic Divisions Only	Omit 2018	Baseline	Include Income 0–138% FPL Sample	New England & Mid-Altantic Divisions Only	Omit 2018
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Treatment	0.0056***	0.0053**	0.0055**	0.0056***	-284.60***	-276.21***	-293.81***	-323.94***	-338.50***	-319.14***	-301.001*	-373.79***
X Post2019	(0.0018)	(0.0023)	(0.0018)	(0.0017)	(35.93)	(49.77)	(112.51)	(43.32)	(48.76)	(62.11)	(114.24)	(85.12)
Sample Size	196,762	273,699	27,319	159,715	196,762	273,699	27,319	159,715	196,762	273,699	27,319	159,715
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
State FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Individual Controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Note: The table reports sensitivity analyses of the estimated effects of the ACA individual mandate repeal on becoming newly uninsured, out-of-pocket premium spending, and medical out-of-pocket spending. For each outcome, the first column reports the baseline specification. The remaining columns sequentially expand the sample to include individuals with household incomes below 138% of the federal poverty line, restrict the sample to states in the New England and Mid-Atlantic Census Divisions, and exclude observations from 2018. Unless otherwise specified, the sample includes adults aged 19 to 64 with household incomes between 138% and 400% of the federal poverty line. All regressions include individual controls, state fixed effects, and year effects. Estimates are weighted using CPS ASEC sampling weights. Standard errors are clustered at the state level and reported in parentheses. ***p<0.01, **p<0.05, *p<0.10.

Table D.4: Leave One Control State Analysis (Source: CPS ASEC, 2015 to 2019)

	Leave Out Massachusetts		Leave Out New Jersey		Leave Out Washington DC	
	Newly Uninsured (1)	Out-of-Premium Spending (2)	Newly Uninsured (3)	Out-of-Premium Spending (4)	Newly Uninsured (5)	Out-of-Premium Spending (6)
Treatment X Post2019	0.0040*	-267.98***	0.0058*	-309.72***	0.0067***	-282.90***
	(0.0023)	(32.65)	(0.0032)	(34.72)	(0.0015)	(36.01)
Sample Size	193,958	193,958	193,414	193,414	194,867	194,867
Pre-2019 Mean	0.021	1,615.54	0.021	1,610.44	0.021	1,616.72
Year FE	✓	✓	✓	✓	✓	✓
State FE	✓	✓	✓	✓	✓	✓
Individual Controls	✓	✓	✓	✓	✓	✓

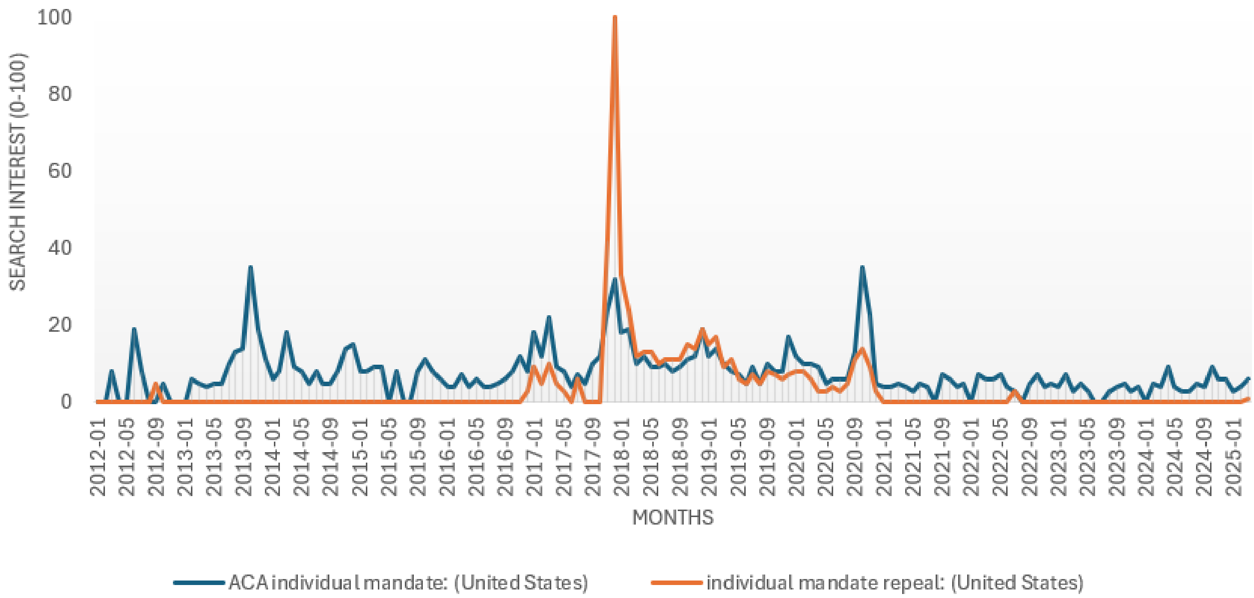
Note: The table reports estimates of the effect of the ACA individual mandate repeal on becoming newly uninsured and out-of-pocket premium spending after sequentially excluding Massachusetts, New Jersey, and the District of Columbia from the control group. The sample includes adults aged 19 to 64 with household incomes between 138% and 400% of the federal poverty line. All regressions include individual controls, state fixed effects, and year effects. Estimates are weighted using CPS ASEC sampling weights. Standard errors are clustered at the state level and reported in parentheses. ***p<0.01, **p<0.05, *p<0.10.

Table D.5: Sensitivity Analysis Using Alternative Treatment-State Samples (Source: CPS ASEC, 2015 to 2019)

	Blue States (CA & NY)		Red States (TX & FL)		Swing States (GA & PA)	
	Newly Uninsured (1)	Out-of-Premium Spending (2)	Newly Uninsured (3)	Out-of-Premium Spending (4)	Newly Uninsured (5)	Out-of-Premium Spending (6)
Treatment X Post2019	0.0079**	-390.81***	0.0059**	-389.18***	0.0042*	-189.11**
	(0.0024)	(50.46)	(0.0015)	(110.46)	(0.0018)	(95.17)
Sample Size	35,474	35,474	30,780	30,780	17,644	17,644
Pre-2019 Mean	0.013	1,410.43	0.020	1,625.50	0.016	1,680.37
Year FE	✓	✓	✓	✓	✓	✓
State FE	✓	✓	✓	✓	✓	✓
Individual Controls	✓	✓	✓	✓	✓	✓

Note: The table reports estimates of the effect of the ACA individual mandate repeal on becoming newly uninsured and out-of-pocket premium spending using alternative subsets of treatment states. Columns 1–2 restrict the treatment group to California and New York, Columns 3–4 to Texas and Florida, and Columns 5–6 to Georgia and Pennsylvania. In each specification, Massachusetts, New Jersey, and the District of Columbia constitute the control group. The sample includes adults aged 19 to 64 with household incomes between 138% and 400% of the federal poverty line. All regressions include individual controls, state fixed effects, and year fixed effects. Estimates are weighted using CPS ASEC sampling weights. Standard errors are clustered at the state level and reported in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.10$.

Figure D.1: Google Trends on Internet Searches related to ACA Individual Mandate and Mandate Repeal (01/2012-01/2025)



Note: Figure D.1 presents U.S. Google search interest for the terms ACA "Individual Mandate" (blue) and "Individual Mandate Repeal" (orange) from January 2012 through January 2025. Search interest is measured using the Google Trends index, which ranges from 0 to 100 and is normalized over the selected period, with 100 representing the peak search interest for each term. Source: Google Trends.